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LAPORAN UJI MINYAK TRAFO

di

UNIT JASA PEMBANGKIT

JERANJANG



Yang dilakukan pada tanggal 28 Juni 2018

Untuk keadaan trafo bulan JUNI 2018

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1.0 Specific Information:

No.	Unit No.	Serial Number	TC	Test Requested
1.	MAIN TRAFO UNIT 1	100691	4	OLTT+ and Corrosive Sulfur
2.	TRAFO 1 ESP UNIT 1	10937	5	OLTT+ and Corrosive Sulfur
3.	TRAFO 2 ESP UNIT 1	10035	6	OLTT+ and Corrosive Sulfur
4.	TRAFO 3 ESP UNIT 1	10031	7	OLTT+ and Corrosive Sulfur
5.	TRAFO 4 ESP UNIT 1	10034	8	OLTT+ and Corrosive Sulfur
6.	MAIN TRAFO UNIT 3	1122170	9	OLTT+ and Corrosive Sulfur
7.	UAT 3	1035082	10	OLTT+ and Corrosive Sulfur
8.	TRAFO 1 ESP UNIT 3	0073084208	11	OLTT+ and Corrosive Sulfur
9.	TRAFO 2 ESP UNIT 3	0070084208	12	OLTT+ and Corrosive Sulfur
10.	TRAFO 3 ESP UNIT 3	0072084208	13	OLTT+ and Corrosive Sulfur
11.	TRAFO 4 UNIT 3	0071084208	14	OLTT+ and Corrosive Sulfur
12.	ASH HANDLING-1	10820108	15	OLTT+ and Corrosive Sulfur
13.	ASH HANDLING-2	10820107	16	OLTT+ and Corrosive Sulfur
14.	TRAFO 1 ESP UNIT 2	10037	17	OLTT+ and Corrosive Sulfur
15.	TRAFO 2 ESP UNIT 2	10036	18	OLTT+ and Corrosive Sulfur
16.	TRAFO 3 ESP UNIT 2	10033	19	OLTT+ and Corrosive Sulfur
17.	TRAFO 4 ESP UNIT 2	10032	20	OLTT+ and Corrosive Sulfur
18.	MAIN TRAFO UNIT 2	1006902	21	OLTT+ and Corrosive Sulfur

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Note:

- * OLTT+ consists of Liquid Screen Tests, Dissolved Gas Analysis (DGA), Water Content, Furan Analysis, Metals-in-Oil Analysis, Liquid Power Factors, and Inhibitor Content (DBPC).

- ** Liquid Screen Test consists of Acidity, Interfacial Tension (IFT), Dielectric, Colour, Specific Gravity, Visualization, and Sediment.

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2.0 Our Comments and Recommendations of The Test Results Analysis:

We had reviewed each laboratory test result against respective benchmark parameter and following are our comments and recommendations.

No.	Unit No.	Serial Number	TC
1.	MAIN TRAFO UNIT 1	100691	4

A) Analysis

- **Acidity** of 0.014mgKOH/g and **IFT** of 41.9mN/m are **ACCEPTABLE**.
- **Dielectric** of 69kV is **ACCEPTABLE**.
- **Specific gravity, Colour, and Visualization** of oil is in the **ACCEPTABLE** range.
- **DGA** shows no further generation of Hydrogen. However, Carbon Monoxide and Carbon Dioxide level is increasing. Although the gas level exceeded the acceptable limit, the CO₂/CO ratio is acceptable, as well as other combustible gasses. For this case, may proceed with annual monitoring.
- **Water Content** in oil of 7ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 0.35% M/dw is graded as **A**. For this transformer, it is recommended to maintain water content in oil at less than 20ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- **Oil Inhibitor** of 0.24% by weight is detected, thus **ACCEPTABLE**.
- Tested on **copper** strip resultant in magenta overcoat on brassy strip, thus **Non-corrosive of 3A**.

B) Recommendation

- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**

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DATE PRINTED 20/07/2018
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - MAIN TRAFU UNIT 1

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
04/07/2017	ONLINE		45	NORMAL		GOOD	
31/07/2017	ONLINE	28	28				
30/08/2017	OFFLINE	49	49	NORMAL	FAIR	GOOD	NONE
29/09/2017	OFFLINE	48		NORMAL	FAIR	GOOD	NONE
30/10/2017	ONLINE	60	64	NORMAL	FAIR	GOOD	NONE
27/11/2017	ONLINE		51				
06/12/2017	OFFLINE	50	48				
22/05/2018	OFFLINE			NORMAL		GOOD	NONE
28/06/2018	ONLINE	50	55				

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 100691
KVA	- 40000
PRI	- 150000 V
SEC	- 6300 V
KILOGRAMS	- 81000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- OFAF
IMPEDENCE	- 13.1%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 30857
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
04/07/2017	ROUTINE	0.017 AC	41.5 AC	58 AC	0.5 AC	0.8691 AC	CLEAR AC	
31/07/2017	ROUTINE	0.017 AC		70 AC	0.5 AC		CLEAR AC	
30/08/2017	ROUTINE	0.017 AC		92 AC	0.5 AC		CLEAR AC	
29/09/2017	ROUTINE	0.014 AC		82 AC	0.5 AC		CLEAR AC	
30/10/2017	ROUTINE	0.014 AC		79 AC	0.5 AC		CLEAR AC	
27/11/2017	ROUTINE	0.014 AC		53 AC	0.5 AC		CLEAR AC	
06/12/2017	ROUTINE	0.011 AC	40.2 AC	87 AC	0.5 AC	0.8745 AC	CLEAR AC	
22/05/2018	ROUTINE	0.011 AC		77 AC	0.5 AC		CLEAR AC	
28/06/2018	ROUTINE	0.014 AC	41.9 AC	69 AC	0.5 AC	0.8716 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	147	9471	43278	32	655	2728	4	5	ND	843	56320
06/12/2017	183	12895	43794	39	715	3166	5	5	ND	947	60802
28/06/2018	176	11044	46776	41	758	3393	5	6	ND	986	62199

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- YES
H2O COOLED	- NO
OIL PUMPS	- YES
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- YES
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

RECOMMENDATION:

NO FURTHER GENERATION OF HYDROGEN IS NOTED. HOWEVER, INCREASING PATTERN OF CARBON OXIDES ARE DETECTED.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
04/07/2017	33	10	0.82	A
31/07/2017	26	8	1.04	B
30/08/2017	34	6	0.49	A
29/09/2017	36	8	0.57	A
30/10/2017	44	10	0.45	A
27/11/2017	42	8	0.40	A
06/12/2017	38	7	0.46	A
22/05/2018	30	10	1.05	B
28/06/2018	41	7	0.35	A

FIELD SERVICE

DATE	SERVICE
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DATE PRINTED 20/07/2018

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - MAIN TRAF0 UNIT 1

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 100691
KVA	- 40000
PRI	- 150000 V
SEC	- 6300 V
KILOGRAMS	- 81000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- OFAF
IMPEDENCE	- 13.1%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 30857
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- YES
H2O COOLED	- NO
OIL PUMPS	- YES
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- YES
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	ND	ND	ND	ND
06/12/2017	ND	ND	ND	ND	ND	ND
28/06/2018	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF
TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	ALUMINIUM	IRON	COPPER
04/07/2017	0.02	0.01	0.01
06/12/2017	ND	0.01	0.04
28/06/2018	ND	ND	0.04

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	0.140 AC
06/12/2017	0.000 AC	0.040 AC
28/06/2018	0.000 AC	0.140 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.24 AC
06/12/2017	0.27 AC
28/06/2018	0.24 AC

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	3A	MAGENTA
06/12/2017	NON-COR	2C	LAVENDER BLUE
28/06/2018	NON-COR	3A	MAGENTA

PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
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FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
2.	TRAFO 1 ESP UNIT 1	10937	5

A) Analysis

- **Acidity** of 0.067mgKOH/g is **QUESTIONABLE** whereas **IFT** of 25.9mN/m is **UNACCEPTABLE**. These two parameters indicate that the oil is contaminated with oxidation byproducts. If left unattended, these oxidation byproducts will eventually deposit on the core of the transformer, causing insufficient heat dissipation, as well as degradation of the paper insulation. Removal of these contaminations is highly recommended.
- **Dielectric** of 32kV is **ACCEPTABLE**.
- **Colour, Specific Gravity, and Visualization** for the oil are within the **ACCEPTABLE** range.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 30ppm is **UNACCEPTABLE**. Meanwhile Water Content in solid insulation of 2.85% M/dw is graded as **E**. For this transformer, it is advisable to maintain the water content level below 20ppm. This indicates high moisture content in the oil. The presence of high moisture content could cause dielectric failure.
- **Furanic compound** 2FOL of 85ppb was detected which indicates the solid insulation is wet. It is important to maintain the oil in a good condition so as not to further deteriorate the paper insulation.
- **Metals-In-Oil** detected were less than 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- No **inhibitor content** is detectable, thus **UNACCEPTABLE**. Lack of inhibitor content will accelerate the oxidation process which produces acid and sludge. These oxidation by-products are depositing and degrading the oil as well as the paper insulation.
- **Corrosive sulfur test** on copper strip resultant in dark orange, thus **Non-corrosive of 1B**.

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B) Recommendation

- It is recommended to perform **Transformer Oil Regeneration** on this unit.
Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts as well as moisture from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

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DATE PRINTED 20/07/2018
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 1 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
04/07/2017		36		NORMAL		GOOD	
31/07/2017	OFFLINE	28					
30/08/2017	OFFLINE	30				GOOD	NONE
29/09/2017	OFFLINE	35		NORMAL		GOOD	NONE
30/10/2017	OFFLINE	50		NORMAL		GOOD	NONE
27/11/2017	ONLINE						
06/12/2017	OFFLINE						
21/05/2018	OFFLINE			NORMAL		GOOD	NONE
28/06/2018	ONLINE						

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10937
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
04/07/2017	ROUTINE	0.056 QU	25.3 UN	50 AC	1.5 AC	0.8529 AC	CLEAR AC	
31/07/2017	ROUTINE	0.059 QU		57 AC	1.5 AC		CLEAR AC	
30/08/2017	ROUTINE	0.056 QU		65 AC	1.5 AC		CLEAR AC	
29/09/2017	ROUTINE	0.059 QU		58 AC	1.5 AC		CLEAR AC	
30/10/2017	ROUTINE	0.059 QU		23 UN	1.5 AC		CLEAR AC	
27/11/2017	ROUTINE	0.062 QU		25 QU	1.5 AC		CLEAR AC	
06/12/2017	ROUTINE	0.062 QU	25.0 UN	48 AC	1.5 AC	0.8583 AC	CLEAR AC	
21/05/2018	ROUTINE	0.065 QU		38 AC	1.5 AC		CLEAR AC	
28/06/2018	ROUTINE	0.067 QU	25.9 UN	32 AC	1.5 AC	0.8560 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	35	20494	67487	4	69	1143	ND	1	ND	109	89233
06/12/2017	39	19933	67198	4	83	1261	ND	1	ND	127	88519
28/06/2018	44	20039	64529	4	119	1696	ND	2	ND	169	86433

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
04/07/2017	26	18	2.34	D
31/07/2017	25	20	2.60	D
30/08/2017	29	18	1.89	D
29/09/2017	32	19	1.80	D
30/10/2017	42	32	1.60	C
27/11/2017	41	33	1.65	C
06/12/2017	30	18	1.89	D
21/05/2018	31	33	3.13	E
28/06/2018	32	30	2.85	E

FIELD SERVICE

DATE	SERVICE
------	---------

DATE PRINTED 20/07/2018

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - TRAFO 1 ESP UNIT 1

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10937
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	95	ND	ND	ND	95
06/12/2017	ND	75	ND	ND	ND	75
28/06/2018	ND	85	ND	ND	ND	85

RECOMMENDATION:
FURAN ANALYSIS SHOWS DETECTION OF 2FOL WHICH INDICATES THAT THE SOLID INSULATION IS WET. IT IS ADVISABLE TO MAINTAIN THE OIL IN A GOOD CONDITION SO AS NOT TO FURTHER DETERIORATE THE PAPER INSULATION.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	ND	0.01	0.01
06/12/2017	ND	ND	ND
28/06/2018	ND	ND	0.03

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	1.870 AC
06/12/2017	0.000 AC	0.570 AC
28/06/2018	0.000 AC	0.680 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	ND UN
06/12/2017	ND UN
28/06/2018	ND UN

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	1B	DARK ORANGE
06/12/2017	NON-COR	1A	LIGHT ORANGE
28/06/2018	NON-COR	1B	DARK ORANGE

®PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
3.	TRAFO 2 ESP UNIT 1	10035	6

A) Analysis

- **Acidity** of 0.059mgKOH/g is **QUESTIONABLE** while **IFT** of 27.6mN/m is **UNACCEPTABLE**. These two parameters indicate there are further accumulation of oxidation byproducts in the oil. Oil oxidation byproducts promotes more acids to be produced by the present acids. Therefore, removal of these contaminations is highly recommended.
- **Dielectric** of 55kV is **ACCEPTABLE**.
- **Colour, Specific Gravity, and Visualization** for the oil are within the **ACCEPTABLE** range.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 17ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.62% M/dw is graded as **C**. For this transformer, it is advisable to maintain the water content level below 20ppm.
- **Furanic compound** 2FOL of 84ppb was detected which indicates the solid insulation is wet. It is important to maintain the oil in a good condition so as not to further deteriorate the paper insulation.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at 25°C and 100°C are **ACCEPTABLE**.
- No **inhibitor content** is detectable, thus **UNACCEPTABLE**. Lack of inhibitor content will accelerate the oxidation process which produces acid and sludge. These oxidation by-products are depositing and degrading the oil as well as the paper insulation.
- **Corrosive sulfur** test on copper strip resultant in dark orange, thus **Non-corrosive of 1B**.

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B) Recommendation

- It is recommended to perform **Transformer Oil Regeneration** on this unit. Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts as well as moisture from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

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DATE PRINTED 20/07/2018
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 2 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
04/07/2017		37		NORMAL		GOOD	
31/07/2017	OFFLINE						
30/08/2017	OFFLINE	39		NORMAL	FAIR	GOOD	NONE
29/09/2017	OFFLINE	40		NORMAL	FAIR	GOOD	NONE
30/10/2017	OFFLINE	44		NORMAL	FAIR	GOOD	NONE
27/11/2017	ONLINE						
06/12/2017	OFFLINE						
21/05/2018	OFFLINE			LOW		GOOD	NONE
28/06/2018	ONLINE						

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10035
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
04/07/2017	ROUTINE	0.042 AC	26.6 UN	41 AC	1.5 AC	0.8540 AC	CLEAR AC	
31/07/2017	ROUTINE	0.045 AC		25 QU	1.5 AC		CLEAR AC	
30/08/2017	ROUTINE	0.042 AC		42 AC	1.5 AC		CLEAR AC	
29/09/2017	ROUTINE	0.048 AC		59 AC	1.5 AC		CLEAR AC	
30/10/2017	ROUTINE	0.051 QU		62 AC	1.5 AC		CLEAR AC	
27/11/2017	ROUTINE	0.051 QU		50 AC	1.5 AC		CLEAR AC	
06/12/2017	ROUTINE	0.053 QU	27.3 UN	51 AC	1.5 AC	0.8589 AC	CLEAR AC	
21/05/2018	ROUTINE	0.056 QU		45 AC	1.5 AC		CLEAR AC	
28/06/2018	ROUTINE	0.059 QU	27.6 UN	55 AC	1.5 AC	0.8570 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	27	18639	66781	3	64	1106	ND	1	ND	95	86621
06/12/2017	20	19360	61590	3	45	728	ND	1	ND	69	81747
28/06/2018	22	22494	64443	4	57	789	ND	ND	ND	83	87809

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
04/07/2017	27	21	2.48	D
31/07/2017	24	27	3.91	E
30/08/2017	28	14	1.65	C
29/09/2017	31	19	1.80	D
30/10/2017	35	17	1.21	B
27/11/2017	35	18	1.28	B
06/12/2017	30	17	1.78	D
21/05/2018	31	20	1.90	D
28/06/2018	32	17	1.62	C

FIELD SERVICE

DATE	SERVICE
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CENNIX LABORATORY SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

6

DATE PRINTED 20/07/2018

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 2 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10035
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	62	ND	ND	ND	62
06/12/2017	ND	67	ND	ND	ND	67
28/06/2018	ND	84	ND	ND	ND	84

RECOMMENDATION:
FURAN ANALYSIS SHOWS DETECTION OF 2FOL WHICH INDICATES THAT THE SOLID INSULATION IS WET. IT IS ADVISABLE TO MAINTAIN THE OIL IN A GOOD CONDITION SO AS NOT TO FURTHER DETERIORATE THE PAPER INSULATION.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	ND	0.01	ND
06/12/2017	ND	ND	0.02
28/06/2018	ND	ND	0.02

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	1.900 AC
06/12/2017	0.000 AC	0.980 AC
28/06/2018	0.000 AC	1.310 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.02 UN
06/12/2017	0.02 UN
28/06/2018	ND UN

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	1A	LIGHT ORANGE
06/12/2017	NON-COR	1B	DARK ORANGE
28/06/2018	NON-COR	1B	DARK ORANGE

®PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
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FIELD SERVICE

DATE	SERVICE
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No.	Unit No.	Serial Number	TC
4.	TRAFO 3 ESP UNIT 1	10031	7

A) Analysis

- **Acidity** of 0.053mgKOH/gm and **IFT** of 29.7mN/m are **QUESTIONABLE**. The levels remained stable without further oil degradation. Stabilized condition may proceed with annual monitoring.
- **Dielectric** of 62kV is **ACCEPTABLE**.
- **Colour, Specific Gravity, and Visualization** for the oil are within the **ACCEPTABLE** range.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 17ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.78% M/dw is graded as **D**. For this transformer, it is advisable to maintain the water content level below 20ppm.
- **Furan analysis** shows minor detection of furanic compounds. This is in line with a healthy solid insulation.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- No **inhibitor content** is detectable, thus **UNACCEPTABLE**. Lack of inhibitor content will accelerate the oxidation process which produces acid and sludge. These oxidation by-products are depositing and degrading the oil as well as the paper insulation.
- **Corrosive sulfur test** on copper strip resultant in dark orange, thus **Non-corrosive of 1B**.

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B) Recommendation

- It is recommended to add oil inhibitor (2,6-DBPC) until the acceptance level (IEEE Std 57-106 suggests 0.09% as minimum level) as soon as possible so as to retard oil oxidation process. By doing so, oil degradation will be slowed down, hence protecting and preserving the transformer insulating material.
- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Do carry-out *Liquid Screen, Liquid Power Factor, Inhibitor, Water Content, Metals-In-Oil, Furan Analysis & DGA periodically for **Condition Monitoring purposes**

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DATE PRINTED 20/07/2018
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 3 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
04/07/2017		28		NORMAL		GOOD	
31/07/2017	ONLINE						
30/08/2017	OFFLINE	35		NORMAL	FAIR	GOOD	NONE
29/09/2017	OFFLINE	38		NORMAL	FAIR	GOOD	NONE
30/10/2017	OFFLINE	44		NORMAL	FAIR	GOOD	NONE
27/11/2017	ONLINE						
06/12/2017	OFFLINE						
21/05/2018	OFFLINE			NORMAL		GOOD	NONE
28/06/2018	ONLINE						

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10031
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
04/07/2017	ROUTINE	0.048 AC	28.4 QU	44 AC	1.0 AC	0.8537 AC	CLEAR AC	
31/07/2017	ROUTINE	0.042 AC		77 AC	1.5 AC		CLEAR AC	
30/08/2017	ROUTINE	0.042 AC		61 AC	1.5 AC		CLEAR AC	
29/09/2017	ROUTINE	0.045 AC		72 AC	1.5 AC		CLEAR AC	
30/10/2017	ROUTINE	0.048 AC		72 AC	1.5 AC		CLEAR AC	
27/11/2017	ROUTINE	0.051 QU		62 AC	1.5 AC		CLEAR AC	
06/12/2017	ROUTINE	0.053 QU	28.3 QU	57 AC	1.5 AC	0.8624 AC	CLEAR AC	
21/05/2018	ROUTINE	0.056 QU		48 AC	1.5 AC		CLEAR AC	
28/06/2018	ROUTINE	0.053 QU	29.7 QU	62 AC	1.5 AC	0.8605 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	32	15273	65111	5	95	1002	ND	2	ND	134	81520
06/12/2017	7	14730	65738	3	42	617	ND	ND	ND	52	81137
28/06/2018	9	17958	66007	4	46	781	ND	1	ND	60	84806

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
04/07/2017	25	32	4.16	E
31/07/2017	25	16	2.08	D
30/08/2017	30	13	1.37	C
29/09/2017	32	14	1.33	B
30/10/2017	37	16	1.04	B
27/11/2017	34	17	1.39	C
06/12/2017	30	16	1.68	C
21/05/2018	31	21	2.00	D
28/06/2018	30	17	1.78	D

FIELD SERVICE

DATE	SERVICE
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CENNIX LABORATORY SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

7

DATE PRINTED 20/07/2018

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 3 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10031
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	32	ND	ND	ND	32
06/12/2017	ND	14	ND	ND	ND	14
28/06/2018	ND	20	ND	ND	ND	20

RECOMMENDATION:
FURAN ANALYSIS SHOWS MINOR DETECTION OF FURANIC COMPOUNDS. THIS IS IN LINE WITH A
HEALTHY SOLID INSULATION.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	0.01	0.01	0.01
06/12/2017	ND	ND	ND
28/06/2018	ND	ND	0.11

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	1.700 AC
06/12/2017	0.000 AC	0.630 AC
28/06/2018	0.000 AC	1.030 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.01 UN
06/12/2017	0.01 UN
28/06/2018	ND UN

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	1A	LIGHT ORANGE
06/12/2017	NON-COR	2C	LAVENDER BLUE
28/06/2018	NON-COR	1B	DARK ORANGE

®PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
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FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
5.	TRAFO 4 ESP UNIT 1	10034	8

A) Analysis

- **Acidity** of 0.037mgKOH/g is **ACCEPTABLE** while **IFT** of 27.5mN/m is **UNACCEPTABLE**. The low IFT level indicates the oil is containing contaminants that had weakened the Interfacial tension of the oil. Removal of these contaminations is highly recommended.
- **Dielectric** of 55kV is **ACCEPTABLE**.
- **Colour, Specific Gravity, and Visualization** for the oil are within the **ACCEPTABLE** range.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 20ppm is **QUESTIONABLE**. Meanwhile Water Content in solid insulation of 2.10% M/dw is graded as **D**. This indicates higher than normal moisture content in the oil. For this transformer, it is advisable to maintain the water content level below 20ppm.
- **Furan analysis** shows minor detection of furanic compounds. This is in line with a healthy solid insulation.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factor** for 25°C and 100°C are **ACCEPTABLE**.
- **Oil Inhibitor** of 0.02% by weight is detected, thus **UNACCEPTABLE**. Insufficient oil inhibitor will cause the insulating oil to oxidize at a faster rate, thus degrading the paper insulation.
- Tested on **copper** strip resultant in magenta overcoat on brassy strip, thus **Non-corrosive of 3A**.

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B) Recommendation

- It is recommended to perform **Transformer Oil Regeneration** on this unit. Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts as well as moisture from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

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DATE PRINTED 20/07/2018
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 4 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
04/07/2017		28		NORMAL		BAD	
31/07/2017	ONLINE						
30/08/2017	OFFLINE	34		NORMAL	FAIR	GOOD	NONE
29/09/2017	OFFLINE	40		NORMAL	FAIR	GOOD	NONE
30/10/2017	OFFLINE	44		NORMAL	FAIR	GOOD	NONE
27/11/2017	ONLINE						
06/12/2017	OFFLINE						
21/05/2018	OFFLINE			LOW	FAIR	GOOD	NONE
28/06/2018	ONLINE						

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10034
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
04/07/2017	ROUTINE	0.028 AC	26.3 UN	24 UN	1.5 AC	0.8596 AC	CLEAR AC	
31/07/2017	ROUTINE	0.022 AC		88 AC	1.5 AC		CLEAR AC	
30/08/2017	ROUTINE	0.028 AC		44 AC	1.5 AC		CLEAR AC	
29/09/2017	ROUTINE	0.031 AC		53 AC	1.0 AC		CLEAR AC	
30/10/2017	ROUTINE	0.028 AC		48 AC	1.5 AC		CLEAR AC	
27/11/2017	ROUTINE	0.034 AC		41 AC	1.5 AC		CLEAR AC	
06/12/2017	ROUTINE	0.031 AC	27.2 UN	41 AC	1.5 AC	0.8645 AC	CLEAR AC	
21/05/2018	ROUTINE	0.034 AC		37 AC	1.5 AC		CLEAR AC	
28/06/2018	ROUTINE	0.037 AC	27.5 UN	55 AC	1.5 AC	0.8635 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	13	12265	66770	5	172	1206	ND	8	ND	198	80439
06/12/2017	8	13331	63527	3	70	720	ND	1	ND	82	77660
28/06/2018	9	17962	64248	4	75	1054	ND	2	ND	90	83354

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
04/07/2017	26	29	3.77	E
31/07/2017	26	15	1.95	D
30/08/2017	30	13	1.37	C
29/09/2017	33	15	1.23	B
30/10/2017	39	18	0.99	B
27/11/2017	37	30	1.95	D
06/12/2017	30	18	1.89	D
21/05/2018	31	23	2.18	D
28/06/2018	30	20	2.10	D

FIELD SERVICE

DATE	SERVICE
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DATE PRINTED 20/07/2018

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 4 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10034
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	18	ND	ND	ND	18
06/12/2017	ND	ND	ND	ND	ND	ND
28/06/2018	ND	21	ND	ND	ND	21

RECOMMENDATION:
FURAN ANALYSIS SHOWS MINOR DETECTION OF FURANIC COMPOUNDS. THIS IS IN LINE WITH A
HEALTHY SOLID INSULATION.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	0.05	0.02	0.01
06/12/2017	ND	ND	0.02
28/06/2018	ND	ND	0.05

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.180 QU	6.680 UN
06/12/2017	0.010 AC	2.180 AC
28/06/2018	0.060 AC	2.850 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.08 UN
06/12/2017	0.06 UN
28/06/2018	0.02 UN

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	2C	LAVENDER BLUE
06/12/2017	NON-COR	2C	LAVENDER BLUE
28/06/2018	NON-COR	3A	MAGENTA

®PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
6.	MAIN TRAF0 UNIT 3	1122170	9

A) Analysis

- **Acidity** of 0.014mgKOH/g and **IFT** of 41.0mN/m are **ACCEPTABLE**.
- **Dielectric** of 85kV is **ACCEPTABLE**.
- **Specific gravity, Colour, and Visualization** of oil is in the **ACCEPTABLE** range.
- **Dissolved Gas Analysis** reveals further increase in Carbon Monoxide and Carbon Dioxide gas levels. Although the gasses level exceeded the acceptable limit, the CO₂/CO ratio is within the tolerable range. For this case, may proceed with annual monitoring. Meanwhile, it is advisable to ascertain the cause of gasses buildup in the transformer.
- **Water Content** in oil of 8ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 0.57% M/dw is graded as **A**. For this transformer, it is recommended to maintain water content in oil at less than 20ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- **Inhibitor content** detected at 0.19% by weight is **QUESTIONABLE**.
Insufficient oil inhibitor will cause the insulating oil to oxidize at a faster rate, thus degrading the paper insulation. Monitor the level annually.
- Tested on **copper** strip resultant in magenta overcoat on brassy strip, thus **Non-corrosive of 3A**.

B) Recommendation

- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**

Page Disclaimer :

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DATE PRINTED 20/07/2018
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - MAIN TRAFU UNIT 3

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
04/07/2017	ONLINE	46	56			GOOD	
31/07/2017	ONLINE	48	50				
30/08/2017	ONLINE	55	60	NORMAL	FAIR	GOOD	NONE
29/09/2017	ONLINE	58	68	NORMAL	FAIR	GOOD	NONE
30/10/2017	OFFLINE	32	32	NORMAL	FAIR	GOOD	NONE
27/11/2017	ONLINE	46	50				
06/12/2017	ONLINE	57	48				
22/05/2018	OFFLINE			NORMAL		GOOD	NONE
28/06/2018	ONLINE	48	50	NORMAL			

NAMEPLATE DATA

MFG	- SHANDONG
DATE MFG	- 2010
MAIN TANK S/N	- 1122170
KVA	- 40000
PRI	- 165000 V
SEC	- 6300 V
KILOGRAMS	- 73920
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAF
IMPEDENCE	- 13.79%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 26708

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
04/07/2017	ROUTINE	0.017 AC	42.3 AC	78 AC	0.5 AC	0.8780 AC	CLEAR AC	
31/07/2017	ROUTINE	0.014 AC		77 AC	0.5 AC		CLEAR AC	
30/08/2017	ROUTINE	0.017 AC		86 AC	0.5 AC		CLEAR AC	
29/09/2017	ROUTINE	0.014 AC		86 AC	0.5 AC		CLEAR AC	
30/10/2017	ROUTINE	0.014 AC		88 AC	0.5 AC		CLEAR AC	
27/11/2017	ROUTINE	0.014 AC		87 AC	0.5 AC		CLEAR AC	
06/12/2017	ROUTINE	0.014 AC	41.5 AC	84 AC	0.5 AC	0.8832 AC	CLEAR AC	
22/05/2018	ROUTINE	0.014 AC		58 AC	0.5 AC		CLEAR AC	
28/06/2018	ROUTINE	0.014 AC	41.0 AC	85 AC	0.5 AC	0.8805 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	24	10131	44084	11	641	2250	1	1	ND	678	57143
06/12/2017	39	14023	43226	11	668	2456	ND	1	ND	719	60424
28/06/2018	18	11431	48283	13	700	2512	1	1	ND	733	62959

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- YES
H2O COOLED	- NO
OIL PUMPS	- YES
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- YES
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

RECOMMENDATION:
CARBON MONOXIDE AND CARBON DIOXIDE LEVEL ARE HIGH. OTHER GASES LEVELS ARE ASSESSED AS ACCEPTABLE.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
04/07/2017	33	15	1.23	B
31/07/2017	35	17	1.21	B
30/08/2017	35	8	0.57	A
29/09/2017	42	10	0.50	A
30/10/2017	28	9	1.06	B
27/11/2017	35	8	0.57	A
06/12/2017	37	9	0.58	A
22/05/2018	31	11	1.04	B
28/06/2018	36	8	0.57	A

FIELD SERVICE

DATE	SERVICE
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CENNIX LABORATORY SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

9

DATE PRINTED 20/07/2018

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **MAIN TRAF0 UNIT 3**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- SHANDONG
DATE MFG	- 2010
MAIN TANK S/N	- 1122170
KVA	- 40000
PRI	- 165000 V
SEC	- 6300 V
KILOGRAMS	- 73920
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAF
IMPEDENCE	- 13.79%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 26708
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- YES
H2O COOLED	- NO
OIL PUMPS	- YES
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- YES
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	ND	ND	ND	ND
06/12/2017	ND	ND	ND	ND	ND	ND
28/06/2018	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF
TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	0.01	0.01	0.02
06/12/2017	0.01	ND	0.03
28/06/2018	ND	0.01	0.06

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	0.130 AC
06/12/2017	0.000 AC	0.070 AC
28/06/2018	0.000 AC	0.110 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.22 AC
06/12/2017	0.22 AC
28/06/2018	0.19 QU

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	3A	MAGENTA
06/12/2017	NON-COR	2C	LAVENDER BLUE
28/06/2018	NON-COR	3A	MAGENTA

®PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
7.	UAT 3	1035082	10

A) Analysis

- **Acidity** of 0.022mgKOH/g and **IFT** of 37.5mN/m are **ACCEPTABLE**.
- **Dielectric** of 79kV is **ACCEPTABLE**.
- **Specific gravity, Colour, and Visualization** of oil is in the **ACCEPTABLE** range.
- **DGA** shows only marginal increases in Carbon Monoxide and Carbon Dioxide levels and could be considered as stable compared to historical values. However, small amount of acetylene is detected. Close monitoring is advised.
- **Water Content** in oil of 10ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 0.82% M/dw is graded as **A**. For this transformer, it is recommended to maintain water content in oil at less than 30ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- **Oil Inhibitor** of 0.21% by weight is detected, thus **ACCEPTABLE**.
- Tested on **copper** strip resultant in magenta overcoat on brassy strip, thus **Non-corrosive of 3A**.

B) Recommendation

- Retest **DGA** in 6 months to confirm if the transformer unit is actively generating the fault gas, thus, warranting any further investigation of the fault detected.
- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Other parameters: Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**

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DATE PRINTED 20/07/2018
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - UAT 3

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
04/07/2017	ONLINE		55			BAD	
31/07/2017	ONLINE						
30/08/2017	ONLINE		58	NORMAL	FAIR	GOOD	NONE
29/09/2017	ONLINE		58	NORMAL	FAIR	GOOD	NONE
30/10/2017	OFFLINE		44	NORMAL	FAIR	GOOD	NONE
27/11/2017	ONLINE		60				
06/12/2017	ONLINE		60				
22/05/2018	ONLINE			NORMAL		GOOD	NONE
28/06/2018	ONLINE		50				

NAMEPLATE DATA

MFG	- SHANDONG
DATE MFG	- 2010
MAIN TANK S/N	- 1035082
KVA	- 6300
PRI	- 6300 V
SEC	- 6300 V
KILOGRAMS	- 14600
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	- 5.78%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 3749

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
04/07/2017	ROUTINE	0.022 AC	38.1 AC	60 AC	0.5 AC	0.8767 AC	CLEAR AC	
31/07/2017	ROUTINE	0.025 AC		88 AC	0.5 AC		CLEAR AC	
30/08/2017	ROUTINE	0.022 AC		89 AC	0.5 AC		CLEAR AC	
29/09/2017	ROUTINE	0.020 AC		80 AC	0.5 AC		CLEAR AC	
30/10/2017	ROUTINE	0.022 AC		86 AC	0.5 AC		CLEAR AC	
27/11/2017	ROUTINE	0.022 AC		86 AC	0.5 AC		CLEAR AC	
06/12/2017	ROUTINE	0.020 AC	37.6 AC	86 AC	0.5 AC	0.8820 AC	CLEAR AC	
22/05/2018	ROUTINE	0.020 AC		81 AC	0.5 AC		CLEAR AC	
28/06/2018	ROUTINE	0.022 AC	37.5 AC	79 AC	0.5 AC	0.8793 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	20	16028	68308	20	971	3536	5	7	ND	1023	88895
06/12/2017	36	18752	66758	19	1116	3622	5	7	ND	1183	90315
28/06/2018	35	13798	68160	22	1144	3644	5	7	1	1214	86816

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

RECOMMENDATION:

RETEST 6 MONTHS: THE DETECTION OF ACETYLENE IS SOME CAUSE FOR CONCERN. CLOSE MONITORING IS NEEDED TO DIAGNOSE THE FAULT

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
04/07/2017	33	11	0.90	A
31/07/2017	32	14	1.33	B
30/08/2017	35	11	0.78	A
29/09/2017	37	12	0.78	A
30/10/2017	29	11	1.15	A
27/11/2017	37	12	0.78	A
06/12/2017	39	12	0.66	A
22/05/2018	29	14	1.47	B
28/06/2018	33	10	0.82	A

FIELD SERVICE

DATE	SERVICE
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DATE PRINTED 20/07/2018

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **UAT 3**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- SHANDONG
DATE MFG	- 2010
MAIN TANK S/N	- 1035082
KVA	- 6300
PRI	- 6300 V
SEC	- 6300 V
KILOGRAMS	- 14600
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	- 5.78%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 3749
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	ND	ND	ND	ND
06/12/2017	ND	ND	ND	ND	ND	ND
28/06/2018	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	0.02	0.01	0.04
06/12/2017	ND	0.01	0.08
28/06/2018	ND	0.01	0.11

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	0.390 AC
06/12/2017	0.000 AC	0.170 AC
28/06/2018	0.000 AC	0.140 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.23 AC
06/12/2017	0.23 AC
28/06/2018	0.21 AC

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	3A	MAGENTA
06/12/2017	NON-COR	2C	LAVENDER BLUE
28/06/2018	NON-COR	3A	MAGENTA

®PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
8.	TRAFO 1 ESP UNIT 3	0073084208	11

A) Analysis

- **Acidity** of 0.022mgKOH/g and **IFT** of 35.0mN/m are **ACCEPTABLE**.
- **Dielectric** of 60kV is **ACCEPTABLE**.
- **Colour** and **Visualization** for the oil are within the **ACCEPTABLE** range.
- **Specific Gravity** value is inside the **QUESTIONABLE** range. Close monitoring is needed and the cause should be identified.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 10ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 0.95% M/dw is graded as **B**. For this transformer, it is advisable to maintain the water content level below 20ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- **Inhibitor content** detected at 0.16% by weight is **QUESTIONABLE**. Insufficient oil inhibitor will cause the insulating oil to oxidize at a faster rate, thus degrading the paper insulation. Monitor the level annually.
- **Corrosive sulfur test** on copper strip resultant in dark orange, thus **Non-corrosive of 1B**.

B) Recommendation

- Monitor the **Specific Gravity** of the oil every 12 months.
- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Other parameters: Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**

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DATE PRINTED 20/07/2018
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 1 ESP UNIT 3**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG - ZHEJIAN
DATE MFG - 2010
MAIN TANK S/N - **0073084208**
KVA -
PRI - 72000 V
SEC - 416 V
KILOGRAMS - 1350
EQUIP TYPE - TRANSFORMER
TRANS CLASS - ONAN
IMPEDENCE -
PHASE/CYCLE - 3/50 Hz
LIQUID TYPE - OIL
LITRES LIQUID - 731

LTC MFG -
LTC S/N -
LTC FAB NO -
LTC MODEL -
LTC LITRES LIQUID -
LTC ENERGIZED -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
31/07/2017	ONLINE						
30/08/2017	ONLINE			NORMAL	FAIR	GOOD	NONE
29/09/2017	ONLINE			NORMAL	FAIR	GOOD	NONE
30/10/2017	OFFLINE			NORMAL	FAIR	GOOD	NONE
28/11/2017	ONLINE						
06/12/2017	ONLINE						
21/05/2018	OFFLINE			NORMAL		GOOD	NONE
28/06/2018	OFFLINE						

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
04/07/2017	ROUTINE	0.022 AC	23.2 UN	32 AC	1.5 AC	0.8432 AC	CLEAR AC	
31/07/2017	ROUTINE	0.028 AC		47 AC	1.5 AC		CLEAR AC	
30/08/2017	ROUTINE	0.028 AC		36 AC	1.5 AC		CLEAR AC	
29/09/2017	ROUTINE	0.025 AC		45 AC	1.5 AC		CLEAR AC	
30/10/2017	ROUTINE	0.025 AC		40 AC	1.5 AC		CLEAR AC	
28/11/2017	ROUTINE	0.025 AC		31 AC	1.5 AC		CLEAR AC	
06/12/2017	ROUTINE	0.022 AC	34.6 AC	31 AC	1.5 AC	0.8180 QU	CLEAR AC	
21/05/2018	ROUTINE	0.022 AC		31 AC	0.5 AC		CLEAR AC	
28/06/2018	ROUTINE	0.022 AC	35.0 AC	60 AC	0.5 AC	0.8154 QU	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	14	12943	68807	3	62	1243	ND	ND	ND	79	83072
06/12/2017	14	12739	65097	3	66	612	ND	ND	ND	83	78531
28/06/2018	19	15552	64448	3	57	1391	ND	ND	ND	79	81470

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

ADDITIONAL EQUIPMENT

RADIATORS - YES
FANS - NO
H2O COOLED - NO
OIL PUMPS - NO
TOP FPV -
BOTTOM FPV -
BUCHOLZ - NO
CONSERVATOR - YES
LTC COMPARTMENT - NO
BUSHING LOC - TOP
FREE BREATHER - YES
HOSE LENGTH -
POWER AVAILABLE - YES

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
04/07/2017	36	26	1.85	D
31/07/2017	31	21	2.00	D
30/08/2017	30	22	2.31	D
29/09/2017	34	20	1.64	C
30/10/2017	28	19	2.24	D
28/11/2017	43	34	1.53	C
06/12/2017	43	31	1.39	C
21/05/2018	31	27	2.57	D
28/06/2018	32	10	0.95	B

FIELD SERVICE

DATE SERVICE

DATE PRINTED 20/07/2018

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 1 ESP UNIT 3**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- ZHEJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 0073084208
KVA	-
PRI	- 72000 V
SEC	- 416 V
KILOGRAMS	- 1350
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 731
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- NO
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	ND	ND	ND	ND
06/12/2017	ND	ND	ND	ND	ND	ND
28/06/2018	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	0.01	0.01	ND
06/12/2017	0.01	ND	ND
28/06/2018	ND	ND	0.02

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	0.350 AC
06/12/2017	0.000 AC	0.040 AC
28/06/2018	0.000 AC	0.050 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	ND UN
06/12/2017	0.18 QU
28/06/2018	0.16 QU

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	1B	DARK ORANGE
06/12/2017	NON-COR	2E	BRASSY GOLD
28/06/2018	NON-COR	1B	DARK ORANGE

®PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
9.	TRAFO 2 ESP UNIT 3	0070084208	12

A) Analysis

- **Acidity** of 0.25mgKOH/g is **ACCEPTABLE** while **IFT** of 30.2mN/m is **QUESTIONABLE**. The levels remained stable without further oil degradation. Stabilized condition may proceed with annual monitoring.
- **Dielectric** of 34kV is **ACCEPTABLE**.
- **Colour** and **Visualization** for the oil are within the **ACCEPTABLE** range.
- **Specific Gravity** value is inside the **QUESTIONABLE** range. Close monitoring is needed and the cause should be identified.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water Content** in oil of 16ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.68% M/dw is graded as **C**. For this transformer, it is recommended to maintain water content in oil at less than 20ppm.
- **Furan analysis** indicates the solid insulation is normal. None of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- **Inhibitor content** detected at 0.19% by weight is **QUESTIONABLE**. Insufficient oil inhibitor will cause the insulating oil to oxidize at a faster rate, thus degrading the paper insulation. Monitor the level annually.
- Tested on **copper** strip resultant in magenta overcoat on brassy strip, thus **Non-corrosive of 3A**.

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B) Recommendation

- Monitor the **Specific Gravity** of the oil every 12 months.
- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Other parameters: Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**

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DATE PRINTED 20/07/2018
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 2 ESP UNIT 3**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
30/08/2017	OFFLINE			NORMAL	FAIR	GOOD	NONE
29/09/2017	ONLINE			NORMAL	FAIR	GOOD	NONE
30/10/2017	OFFLINE			NORMAL	FAIR	GOOD	NONE
28/11/2017	ONLINE						
06/12/2017	ONLINE						
21/05/2018	OFFLINE			NORMAL		GOOD	NONE
28/06/2018	OFFLINE						

NAMEPLATE DATA

MFG	- ZHEJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 0070084208
KVA	-
PRI	- 72000 V
SEC	- 415 V
KILOGRAMS	- 1350
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 731
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
04/07/2017	ROUTINE	0.118 UN	16.2 UN	24 UN	3.0 AC	0.8435 AC	CLEAR AC	
31/07/2017	ROUTINE	0.112 UN		28 QU	3.0 AC		CLEAR AC	
30/08/2017	ROUTINE	0.109 UN		34 AC	2.5 AC		CLEAR AC	
29/09/2017	ROUTINE	0.115 UN		21 UN	2.5 AC		CLEAR AC	
30/10/2017	ROUTINE	0.118 UN		33 AC	2.5 AC		CLEAR AC	
28/11/2017	ROUTINE	0.017 AC		25 QU	0.5 AC		CLEAR AC	
06/12/2017	ROUTINE	0.022 AC	29.7 QU	29 QU	0.5 AC	0.8106 QU	CLEAR AC	
21/05/2018	ROUTINE	0.022 AC		28 QU	0.5 AC		CLEAR AC	
28/06/2018	ROUTINE	0.025 AC	30.2 QU	34 AC	0.5 AC	0.8080 QU	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	38	16781	70523	5	129	2431	3	5	ND	180	89915
06/12/2017	42	19523	66059	3	102	912	ND	ND	ND	147	86641
28/06/2018	43	22272	68719	4	38	1226	ND	ND	ND	85	92302

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
04/07/2017	37	39	2.54	D
31/07/2017	31	38	3.61	E
30/08/2017	26	17	2.21	D
29/09/2017	36	37	2.63	D
30/10/2017	28	20	2.36	D
28/11/2017	43	28	1.26	B
06/12/2017	43	44	1.98	D
21/05/2018	31	30	2.85	E
28/06/2018	30	16	1.68	C

FIELD SERVICE

DATE	SERVICE
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DATE PRINTED 20/07/2018

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 2 ESP UNIT 3**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- ZHEJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 0070084208
KVA	-
PRI	- 72000 V
SEC	- 415 V
KILOGRAMS	- 1350
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 731
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	169	ND	ND	169
06/12/2017	ND	ND	ND	ND	ND	ND
28/06/2018	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	ND	0.01	0.01
06/12/2017	ND	ND	ND
28/06/2018	ND	ND	0.02

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	1.330 AC
06/12/2017	0.000 AC	0.030 AC
28/06/2018	0.000 AC	0.040 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	ND UN
06/12/2017	0.20 AC
28/06/2018	0.19 QU

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	2C	LAVENDER BLUE
06/12/2017	NON-COR	3A	MAGENTA
28/06/2018	NON-COR	3A	MAGENTA

®PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
10.	TRAFO 3 ESP UNIT 3	0072084208	13

A) Analysis

- **Acidity** of 0.017mgKOH/g and **IFT** of 38.4mN/m are **ACCEPTABLE**.
- **Dielectric** of 43kV is **ACCEPTABLE**.
- **Colour** and **Visualization** for the oil are within the **ACCEPTABLE** range.
- **Specific Gravity** value is inside the **QUESTIONABLE** range. Close monitoring is needed and the cause should be identified.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 22ppm is **QUESTIONABLE**. Meanwhile Water Content in solid insulation of 1.56% M/dw is graded as **C**. For this transformer, it is advisable to maintain the water content level below 20ppm. This indicates higher than normal moisture content in the oil.
- **Furan analysis** shows minor detection of furanic compounds. This is in line with a healthy solid insulation.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- **Oil Inhibitor** of 0.20% by weight is detected, thus **ACCEPTABLE**.
- **Corrosive sulfur** test on copper strip resultant in colour changes to brassy and gold, thus **Non-corrosive of 2E**.

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B) Recommendation

- Monitor the **Specific Gravity** of the oil every 12 months.
- Retest **Moisture** and **Dielectric** in 6 months to assess if corrective work on the transformer oil is required.
- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Other parameters: Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**

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DATE PRINTED 20/07/2018
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 3 ESP UNIT 3**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
04/07/2017	ONLINE						
31/07/2017	ONLINE						
30/08/2017	ONLINE			NORMAL	FAIR	GOOD	NONE
29/09/2017	ONLINE			NORMAL	FAIR	GOOD	NONE
30/10/2017	OFFLINE			LOW	FAIR	GOOD	NONE
28/11/2017	ONLINE						
06/12/2017	ONLINE						
21/05/2018	OFFLINE			NORMAL		GOOD	NONE
28/06/2018	ONLINE						

NAMEPLATE DATA

MFG	- ZHEJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 0072084208
KVA	-
PRI	- 72000 V
SEC	- 415 V
KILOGRAMS	- 1350
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 731
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
04/07/2017	ROUTINE	0.025 AC	27.8 UN	40 AC	1.5 AC	0.8415 AC	CLEAR AC	
31/07/2017	ROUTINE	0.025 AC		66 AC	1.5 AC		CLEAR AC	
30/08/2017	ROUTINE	0.025 AC		50 AC	1.5 AC		CLEAR AC	
29/09/2017	ROUTINE	0.028 AC		59 AC	1.5 AC		CLEAR AC	
30/10/2017	ROUTINE	0.028 AC		55 AC	1.5 AC		CLEAR AC	
28/11/2017	ROUTINE	0.014 AC		39 AC	0.5 AC		CLEAR AC	
06/12/2017	ROUTINE	0.014 AC	39.8 AC	45 AC	0.5 AC	0.8099 QU	CLEAR AC	
21/05/2018	ROUTINE	0.014 AC		54 AC	0.5 AC		CLEAR AC	
28/06/2018	ROUTINE	0.017 AC	38.4 AC	43 AC	0.5 AC	0.8075 QU	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	21	16364	70694	3	38	1068	ND	ND	ND	62	88188
06/12/2017	15	14515	65277	2	40	559	ND	ND	ND	57	80408
28/06/2018	30	15202	64188	4	65	1833	ND	ND	ND	99	81322

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
04/07/2017	37	27	1.75	D
31/07/2017	29	18	1.89	D
30/08/2017	35	21	1.49	C
29/09/2017	37	22	1.43	C
30/10/2017	28	16	1.89	D
28/11/2017	44	24	1.08	B
06/12/2017	44	29	1.30	B
21/05/2018	32	23	2.18	D
28/06/2018	35	22	1.56	C

FIELD SERVICE

DATE	SERVICE
------	---------

DATE PRINTED 20/07/2018

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 3 ESP UNIT 3**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- ZHEJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 0072084208
KVA	-
PRI	- 72000 V
SEC	- 415 V
KILOGRAMS	- 1350
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 731
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	ND	ND	ND	ND
06/12/2017	ND	ND	ND	ND	ND	ND
28/06/2018	ND	4	ND	ND	ND	4

RECOMMENDATION:
FURAN ANALYSIS SHOWS MINOR DETECTION OF FURANIC COMPOUNDS. THIS IS IN LINE WITH A
HEALTHY SOLID INSULATION.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	ND	0.01	ND
06/12/2017	ND	ND	ND
28/06/2018	ND	ND	0.02

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	0.620 AC
06/12/2017	0.000 AC	0.080 AC
28/06/2018	0.000 AC	0.010 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.01 UN
06/12/2017	0.21 AC
28/06/2018	0.20 AC

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	2C	LAVENDER BLUE
06/12/2017	NON-COR	2E	BRASSY GOLD
28/06/2018	NON-COR	2E	BRASSY GOLD

®PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
11.	TRAFO 4 UNIT 3	0071084208	14

A) Analysis

- **Acidity** of 0.025mgKOH/g is **ACCEPTABLE** while **IFT** of 30.5mN/m is **QUESTIONABLE**. The levels remained stable without further oil degradation. Stabilized condition may proceed with annual monitoring.
- **Dielectric** of 71kV is **ACCEPTABLE**.
- **Colour, Specific Gravity, and Visualization** for the oil are within the **ACCEPTABLE** range.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 11ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 0.78% M/dw is graded as **A**. For this transformer, it is advisable to maintain the water content level below 20ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- No **inhibitor content** is detectable, thus **UNACCEPTABLE**. Lack of inhibitor content will accelerate the oxidation process which produces acid and sludge. These oxidation by-products are depositing and degrading the oil as well as the paper insulation.
- **Corrosive sulfur** test on copper strip resultant in dark orange, thus **Non-corrosive of 1B**.

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B) Recommendation

- It is recommended to add oil inhibitor (2,6-DBPC) until the acceptance level (IEEE Std 57-106 suggests 0.09% as minimum level) as soon as possible so as to retard oil oxidation process. By doing so, oil degradation will be slowed down, hence protecting and preserving the transformer insulating material.
- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Do carry-out *Liquid Screen, Liquid Power Factor, Inhibitor, Water Content, Metals-In-Oil, Furan Analysis & DGA periodically for **Condition Monitoring purposes**

Page Disclaimer :

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DATE PRINTED 20/07/2018
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 4 UNIT 3**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
30/08/2017	OFFLINE			NORMAL	FAIR	GOOD	NONE
29/09/2017	OFFLINE			NORMAL	FAIR	GOOD	NONE
30/10/2017	OFFLINE			NORMAL	FAIR	GOOD	NONE
28/11/2017	OFFLINE						
06/12/2017	OFFLINE						
21/05/2018	OFFLINE			NORMAL		GOOD	NONE
28/06/2018	ONLINE						

NAMEPLATE DATA

MFG	- ZHEJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 0071084208
KVA	-
PRI	- 72000 V
SEC	- 415 V
KILOGRAMS	- 1350
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 731
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
04/07/2017	ROUTINE	0.020 AC	31.2 QU	62 AC	1.0 AC	0.8420 AC	CLEAR AC	
04/07/2017	ROUTINE	0.022 AC		60 AC	1.0 AC		CLEAR AC	
30/08/2017	ROUTINE	0.020 AC		55 AC	1.0 AC		CLEAR AC	
29/09/2017	ROUTINE	0.022 AC		59 AC	1.0 AC		CLEAR AC	
30/10/2017	ROUTINE	0.025 AC		55 AC	1.0 AC		CLEAR AC	
28/11/2017	ROUTINE	0.022 AC		69 AC	1.0 AC		CLEAR AC	
06/12/2017	ROUTINE	0.025 AC	29.8 QU	43 AC	1.0 AC	0.8471 AC	CLEAR AC	
21/05/2018	ROUTINE	0.028 AC		37 AC	1.0 AC		CLEAR AC	
28/06/2018	ROUTINE	0.025 AC	30.5 QU	71 AC	1.0 AC	0.8442 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	10	18669	66036	3	24	665	ND	ND	ND	37	85407
06/12/2017	14	15414	63712	2	22	675	ND	ND	ND	38	79839
28/06/2018	20	14690	67822	2	22	683	ND	ND	ND	44	83239

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
04/07/2017	32	19	1.80	D
04/07/2017	31	15	1.42	C
30/08/2017	27	17	2.01	D
29/09/2017	29	15	1.58	C
30/10/2017	28	15	1.77	D
28/11/2017	29	16	1.68	C
06/12/2017	32	19	1.80	D
21/05/2018	30	21	2.20	D
28/06/2018	36	11	0.78	A

FIELD SERVICE

DATE	SERVICE
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DATE PRINTED 20/07/2018

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 4 UNIT 3**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- ZHEJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 0071084208
KVA	-
PRI	- 72000 V
SEC	- 415 V
KILOGRAMS	- 1350
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 731
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	ND	ND	ND	ND
06/12/2017	ND	ND	ND	ND	ND	ND
28/06/2018	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF
TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	ND	ND	ND
06/12/2017	ND	0.03	ND
28/06/2018	ND	0.03	0.02

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	1.360 AC
06/12/2017	0.000 AC	0.370 AC
28/06/2018	0.000 AC	0.550 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.01 UN
06/12/2017	0.03 UN
28/06/2018	ND UN

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	3A	MAGENTA
06/12/2017	NON-COR	1B	DARK ORANGE
28/06/2018	NON-COR	1B	DARK ORANGE

®PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
12.	ASH HANDLING-1	10820108	15

A) Analysis

- **Acidity** of 0.025mgKOH/g and **IFT** of 37.5mN/m are **ACCEPTABLE**.
- **Dielectric** of 42kV is **ACCEPTABLE**.
- **Specific gravity, Colour, and Visualization** of oil is in the **ACCEPTABLE** range.
- **Dissolved Gas Analysis** shows Carbon Monoxide and Carbon Dioxide levels to have stabilized. Other gases remain within normal limits and does not indicate any fault afflicting the transformer unit. Proceed with annual monitoring.
- **Water Content** in oil of 24ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 2.28% M/dw is graded as **C**. For this transformer, it is recommended to maintain water content in oil at less than 30ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- **Oil Inhibitor** of 0.22% by weight is detected, thus **ACCEPTABLE**.
- **Corrosive sulfur test** on copper strip resultant in dark orange, thus **Non-corrosive of 1B**.

B) Recommendation

- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**

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DATE PRINTED 20/07/2018
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **ASH HANDLING-1**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
04/07/2017	ONLINE	46					
31/07/2017	ONLINE						
30/08/2017	OFFLINE	30		NORMAL	FAIR		NONE
29/09/2017	OFFLINE	40		NORMAL	FAIR		NONE
30/10/2017	ONLINE	45		NORMAL	FAIR		NONE
27/11/2017	ONLINE	46					
06/12/2017	ONLINE	52					
22/05/2018	ONLINE	45		NORMAL		GOOD	NONE
28/06/2018	ONLINE						

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 10820108
KVA	- 1600
PRI	- 6300 V
SEC	- 400 V
KILOGRAMS	- 4505
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	- 5.56%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 1023

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
04/07/2017	ROUTINE	0.022 AC	38.1 AC	38 AC	1.0 AC	0.8779 AC	CLEAR AC	
31/07/2017	ROUTINE	0.025 AC		56 AC	1.0 AC		CLEAR AC	
30/08/2017	ROUTINE	0.022 AC		53 AC	1.0 AC		CLEAR AC	
29/09/2017	ROUTINE	0.022 AC		57 AC	1.0 AC		CLEAR AC	
30/10/2017	ROUTINE	0.022 AC		45 AC	1.0 AC		CLEAR AC	
27/11/2017	ROUTINE	0.020 AC		36 AC	1.0 AC		CLEAR AC	
06/12/2017	ROUTINE	0.017 AC	37.5 AC	34 AC	1.0 AC	0.8832 AC	CLEAR AC	
22/05/2018	ROUTINE	0.022 AC		33 AC	1.0 AC		CLEAR AC	
28/06/2018	ROUTINE	0.025 AC	37.5 AC	42 AC	1.0 AC	0.8804 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	35	10984	71058	20	441	5037	7	8	ND	511	87590
06/12/2017	49	12034	67124	20	558	5440	8	8	ND	643	85241
28/06/2018	53	10939	69394	20	557	5425	9	8	ND	647	86405

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

RECOMMENDATION:
GAS LEVELS HAVE REMAINED STABLE. PROCEED WITH ANNUAL MONITORING.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
04/07/2017	34	28	2.30	C
31/07/2017	26	21	2.73	D
30/08/2017	28	16	1.89	B
29/09/2017	34	16	1.31	B
30/10/2017	33	20	1.64	B
27/11/2017	36	29	2.06	C
06/12/2017	36	26	1.85	B
22/05/2018	33	26	2.13	C
28/06/2018	32	24	2.28	C

FIELD SERVICE

DATE	SERVICE
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DATE PRINTED 20/07/2018

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **ASH HANDLING-1**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 10820108
KVA	- 1600
PRI	- 6300 V
SEC	- 400 V
KILOGRAMS	- 4505
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	- 5.56%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 1023
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	ND	ND	ND	ND
06/12/2017	ND	ND	ND	ND	ND	ND
28/06/2018	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF
TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	ND	0.01	0.01
06/12/2017	ND	ND	0.01
28/06/2018	ND	0.01	0.04

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	0.730 AC
06/12/2017	0.000 AC	0.240 AC
28/06/2018	0.000 AC	0.460 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.23 AC
06/12/2017	0.24 AC
28/06/2018	0.22 AC

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	1B	DARK ORANGE
06/12/2017	NON-COR	2C	LAVENDER BLUE
28/06/2018	NON-COR	1B	DARK ORANGE

®PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
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FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
13.	ASH HANDLING-2	10820107	16

A) Analysis

- **Acidity** of 0.022mgKOH/g and **IFT** of 35.0mN/m are **ACCEPTABLE**.
- **Dielectric** of 50kV is **ACCEPTABLE**.
- **Specific gravity, Colour, and Visualization** of oil is in the **ACCEPTABLE** range.
- **Dissolved Gas Analysis** of this sample shows Carbon Dioxide detected increasing. Other gases are assessed as acceptable. The CO₂/CO ratio is beyond the tolerable limit. Close monitoring is needed.
- **Water Content** in oil of 16ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.89% M/dw is graded as **B**. For this transformer, it is recommended to maintain water content in oil at less than 30ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- **Oil Inhibitor** of 0.20% by weight is detected, thus **ACCEPTABLE**.
- **Corrosive sulfur** test on copper strip resultant in magenta, thus **Non-corrosive of 3A**.

B) Recommendation

- Retest **DGA** in 6 months to monitor the Carbon Dioxide build up in the unit.
- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Other parameters: Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**.

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DATE PRINTED 20/07/2018
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **ASH HANDLING-2**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
04/07/2017	ONLINE		40				
31/07/2017	ONLINE						
30/08/2017	ONLINE	47		NORMAL	FAIR		NONE
29/09/2017	ONLINE	60		NORMAL	FAIR		NONE
30/10/2017	OFFLINE	32		NORMAL	FAIR		NONE
27/11/2017	OFFLINE	40					
06/12/2017	ONLINE	48					
22/05/2018	ONLINE	45				GOOD	NONE
28/06/2018	OFFLINE						

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 10820107
KVA	- 1600
PRI	- 6300 V
SEC	- 400 V
KILOGRAMS	- 4505
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	- 5.6%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 1023
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
04/07/2017	ROUTINE	0.022 AC	37.3 AC	36 AC	1.0 AC	0.8788 AC	CLEAR AC	
31/07/2017	ROUTINE	0.022 AC		40 AC	1.0 AC		CLEAR AC	
30/08/2017	ROUTINE	0.022 AC		31 AC	1.5 AC		CLEAR AC	
29/09/2017	ROUTINE	0.022 AC		28 QU	1.5 AC		CLEAR AC	
30/10/2017	ROUTINE	0.020 AC		46 AC	1.5 AC		CLEAR AC	
27/11/2017	ROUTINE	0.022 AC		47 AC	1.5 AC		CLEAR AC	
06/12/2017	ROUTINE	0.020 AC	36.6 AC	24 UN	1.5 AC	0.8840 AC	CLEAR AC	
22/05/2018	ROUTINE	0.020 AC		24 UN	1.5 AC		CLEAR AC	
28/06/2018	ROUTINE	0.022 AC	35.0 AC	50 AC	1.5 AC	0.8811 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	18	21157	64224	4	129	1676	ND	1	ND	152	87209
06/12/2017	20	15791	63530	5	275	2783	ND	4	ND	304	82408
28/06/2018	32	22904	67139	4	249	3112	ND	5	ND	290	93445

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- NO
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- NO
HOSE LENGTH	-
POWER AVAILABLE	- YES

RECOMMENDATION:
CARBON DIOXIDE GAS LEVEL IS HIGH.OTHER GASES ARE ASSESSED AS ACCEPTABLE.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
04/07/2017	29	21	2.20	C
31/07/2017	31	32	3.04	D
30/08/2017	34	29	2.38	C
29/09/2017	40	31	1.71	B
30/10/2017	29	21	2.20	C
27/11/2017	29	20	2.10	C
06/12/2017	34	32	2.62	D
22/05/2018	33	32	2.62	D
28/06/2018	28	16	1.89	B

FIELD SERVICE

DATE	SERVICE
------	---------

DATE PRINTED 20/07/2018

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **ASH HANDLING-2**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 10820107
KVA	- 1600
PRI	- 6300 V
SEC	- 400 V
KILOGRAMS	- 4505
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	- 5.6%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 1023
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- NO
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- NO
HOSE LENGTH	-
POWER AVAILABLE	- YES

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	ND	ND	ND	ND
06/12/2017	ND	ND	3	ND	ND	3
28/06/2018	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	ND	0.01	ND
06/12/2017	ND	0.01	0.01
28/06/2018	ND	ND	0.02

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	1.350 AC
06/12/2017	0.000 AC	0.270 AC
28/06/2018	0.000 AC	0.490 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.22 AC
06/12/2017	0.24 AC
28/06/2018	0.20 AC

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	3A	MAGENTA
06/12/2017	NON-COR	1A	LIGHT ORANGE
28/06/2018	NON-COR	3A	MAGENTA

®PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
14.	TRAFO 1 ESP UNIT 2	10037	17

A) Analysis

- **Acidity** of 0.025mgKOH/g and **IFT** of 32.9mN/m are **ACCEPTABLE**.
- **Dielectric** of 77kV is **ACCEPTABLE**.
- **Colour, Specific Gravity, and Visualization** for the oil are within the **ACCEPTABLE** range.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 11ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.30% M/dw is graded as **B**. For this transformer, it is advisable to maintain the water content level below 20ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- No **inhibitor content** is detectable, thus **UNACCEPTABLE**. Lack of inhibitor content will accelerate the oxidation process which produces acid and sludge. These oxidation by-products are depositing and degrading the oil as well as the paper insulation.
- **Corrosive sulfur** test on copper strip resultant in lavender blue, thus **Non-corrosive of 2C**.

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B) Recommendation

- It is recommended to add oil inhibitor (2,6-DBPC) until the acceptance level (IEEE Std 57-106 suggests 0.09% as minimum level) as soon as possible so as to retard oil oxidation process. By doing so, oil degradation will be slowed down, hence protecting and preserving the transformer insulating material.
- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Do carry-out *Liquid Screen, Liquid Power Factor, Inhibitor, Water Content, Metals-In-Oil, Furan Analysis & DGA periodically for **Condition Monitoring purposes**

Page Disclaimer :

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DATE PRINTED 20/07/2018
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 1 ESP UNIT 2**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
28/11/2017	OFFLINE						
06/12/2017	OFFLINE						
21/05/2018	OFFLINE			NORMAL		GOOD	NONE
28/06/2018	OFFLINE						

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10037
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	-
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
28/11/2017	ROUTINE	0.020 AC		48 AC	1.0 AC		CLEAR AC	
06/12/2017	ROUTINE	0.022 AC		22 UN	1.0 AC		CLEAR AC	
21/05/2018	ROUTINE	0.025 AC		24 UN	1.0 AC		CLEAR AC	
28/06/2018	ROUTINE	0.025 AC	32.9 AC	77 AC	1.0 AC	0.8693 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
28/06/2018	93	17225	43231	5	57	775	ND	2	ND	157	61388

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
28/11/2017	27	21	2.48	D
06/12/2017	27	35	4.13	E
21/05/2018	31	32	3.04	E
28/06/2018	28	11	1.30	B

FIELD SERVICE
DATE SERVICE

DATE PRINTED 20/07/2018

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 1 ESP UNIT 2**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10037
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	-
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
28/06/2018	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
28/06/2018	ND	0.06	0.03

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
28/06/2018	0.030 AC	0.420 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
28/06/2018	ND UN

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
28/06/2018	NON-COR	2C	LAVENDER BLUE

®PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
15.	TRAFO 2 ESP UNIT 2	10036	18

A) Analysis

- **Acidity** of 0.056mgKOH/g and **IFT** of 31.3mN/m are **QUESTIONABLE**. The levels remained stable without further oil degradation. Stabilized condition may proceed with annual monitoring.
- **Dielectric** of 66kV is **ACCEPTABLE**.
- **Colour, Specific Gravity, and Visualization** for the oil are within the **ACCEPTABLE** range.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 13ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.53% M/dw is graded as **C**. For this transformer, it is advisable to maintain the water content level below 20ppm.
- **Furan analysis** shows minor detection of furanic compounds. This is in line with a healthy solid insulation.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- No **inhibitor content** is detectable, thus **UNACCEPTABLE**. Lack of inhibitor content will accelerate the oxidation process which produces acid and sludge. These oxidation by-products are depositing and degrading the oil as well as the paper insulation.
- **Corrosive sulfur** test on copper strip resultant in magenta, thus **Non-corrosive of 3A**.

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B) Recommendation

- It is recommended to add oil inhibitor (2,6-DBPC) until the acceptance level (IEEE Std 57-106 suggests 0.09% as minimum level) as soon as possible so as to retard oil oxidation process. By doing so, oil degradation will be slowed down, hence protecting and preserving the transformer insulating material.
- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Do carry-out *Liquid Screen, Liquid Power Factor, Inhibitor, Water Content, Metals-In-Oil, Furan Analysis & DGA periodically for **Condition Monitoring purposes**

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DATE PRINTED 20/07/2018
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 2 ESP UNIT 2**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
28/11/2017	OFFLINE						
06/12/2017	OFFLINE						
21/05/2018	OFFLINE			NORMAL		GOOD	NONE
28/06/2018	ONLINE						

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10036
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
28/11/2017	ROUTINE	0.048 AC		55 AC	1.0 AC		CLEAR AC	
06/12/2017	ROUTINE	0.051 QU		59 AC	1.0 AC		CLEAR AC	
21/05/2018	ROUTINE	0.053 QU		55 AC	1.0 AC		CLEAR AC	
28/06/2018	ROUTINE	0.056 QU	31.3 QU	66 AC	1.0 AC	0.8573 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
28/06/2018	8	22714	64906	5	67	879	ND	ND	ND	80	88579

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
28/11/2017	28	13	1.53	C
06/12/2017	28	14	1.65	C
21/05/2018	31	16	1.52	C
28/06/2018	27	13	1.53	C

FIELD SERVICE

DATE	SERVICE
------	---------

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

DATE PRINTED 20/07/2018

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 2 ESP UNIT 2**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10036
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
28/06/2018	ND	42	ND	ND	ND	42

RECOMMENDATION:
FURAN ANALYSIS SHOWS MINOR DETECTION OF FURANIC COMPOUNDS. THIS IS IN LINE WITH A
HEALTHY SOLID INSULATION.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
28/06/2018	ND	ND	0.02

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
28/06/2018	0.000 AC	0.680 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
28/06/2018	ND UN

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
28/06/2018	NON-COR	3A	MAGENTA

®PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
16.	TRAFO 3 ESP UNIT 2	10033	19

A) Analysis

- **Acidity** of 0.062mgKOH/g is **QUESTIONABLE** while **IFT** of 26.3mN/m is **UNACCEPTABLE**. These two parameters indicate there are further accumulation of oxidation byproducts in the oil. Oil oxidation byproducts promotes more acids to be produced by the present acids. Therefore, removal of these contaminations is highly recommended.
- **Dielectric** of 78kV is **ACCEPTABLE**.
- **Colour, Specific Gravity, and Visualization** for the oil are within the **ACCEPTABLE** range.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 14ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.65% M/dw is graded as **C**. For this transformer, it is advisable to maintain the water content level below 20ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at 25°C and 100°C are **ACCEPTABLE**.
- No **inhibitor content** is detectable, thus **UNACCEPTABLE**. Lack of inhibitor content will accelerate the oxidation process which produces acid and sludge. These oxidation by-products are depositing and degrading the oil as well as the paper insulation.
- **Corrosive sulfur** test on copper strip resultant in magenta, thus **Non-corrosive of 3A**.

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B) Recommendation

- It is recommended to perform **Transformer Oil Regeneration** on this unit.
Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

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DATE PRINTED 20/07/2018
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 3 ESP UNIT 2**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
28/11/2017	OFFLINE						
06/12/2017	OFFLINE						
21/05/2018	OFFLINE			NORMAL		GOOD	NONE
28/06/2018	ONLINE						

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10033
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
28/11/2017	ROUTINE	0.053 QU		42 AC	1.5 AC		CLEAR AC	
06/12/2017	ROUTINE	0.056 QU		60 AC	1.5 AC		CLEAR AC	
21/05/2018	ROUTINE	0.059 QU		51 AC	1.5 AC		CLEAR AC	
28/06/2018	ROUTINE	0.062 QU	26.3 UN	78 AC	1.5 AC	0.8602 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
28/06/2018	14	17627	48672	4	47	935	ND	ND	ND	65	67299

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
28/11/2017	28	15	1.77	D
06/12/2017	28	16	1.89	D
21/05/2018	31	18	1.71	D
28/06/2018	28	14	1.65	C

FIELD SERVICE

DATE	SERVICE
------	---------

DATE PRINTED 20/07/2018

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - TRAF0 3 ESP UNIT 2

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10033
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
28/06/2018	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF
TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	ALUMINIUM	IRON	COPPER
28/06/2018	ND	ND	0.02

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
28/06/2018	0.000 AC	0.570 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
28/06/2018	ND UN

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
28/06/2018	NON-COR	3A	MAGENTA

PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
17.	TRAFO 4 ESP UNIT 2	10032	20

A) Analysis

- **Acidity** of 0.028mgKOH/g and **IFT** of 32.6mN/m are **ACCEPTABLE**.
- **Dielectric** of 88kV is **ACCEPTABLE**.
- **Colour, Specific Gravity, and Visualization** for the oil are within the **ACCEPTABLE** range.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 15ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.77% M/dw is graded as **D**. For this transformer, it is advisable to maintain the water content level below 20ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- **Oil Inhibitor** of 0.01% by weight is detected, thus **UNACCEPTABLE**. Lack of inhibitor content will accelerate the oxidation process which produces acid and sludge. These oxidation by-products are depositing and degrading the oil as well as the paper insulation.
- **Corrosive sulfur** test on copper strip resultant in lavender blue, thus **Non-corrosive of 2C**.

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B) Recommendation

- It is recommended to add oil inhibitor (2,6-DBPC) until the acceptance level (IEEE Std 57-106 suggests 0.09% as minimum level) as soon as possible so as to retard oil oxidation process. By doing so, oil degradation will be slowed down, hence protecting and preserving the transformer insulating material.
- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Do carry-out *Liquid Screen, Liquid Power Factor, Inhibitor, Water Content, Metals-In-Oil, Furan Analysis & DGA periodically for **Condition Monitoring purposes**

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DATE PRINTED 20/07/2018
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 4 ESP UNIT 2**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
28/11/2017	OFFLINE						
06/12/2017	OFFLINE						
21/05/2018	OFFLINE					GOOD	NONE
28/06/2018	ONLINE						

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10032
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
28/11/2017	ROUTINE	0.020 AC		62 AC	1.0 AC		CLEAR AC	
06/12/2017	ROUTINE	0.022 AC		58 AC	1.0 AC		CLEAR AC	
21/05/2018	ROUTINE	0.025 AC		40 AC	1.0 AC		CLEAR AC	
28/06/2018	ROUTINE	0.028 AC	32.6 AC	88 AC	1.0 AC	0.8648 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
28/06/2018	22	25201	74464	5	85	823	ND	ND	ND	112	100600

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
28/11/2017	28	16	1.89	D
06/12/2017	28	17	2.01	D
21/05/2018	31	18	1.71	D
28/06/2018	27	15	1.77	D

FIELD SERVICE
DATE SERVICE

DATE PRINTED 20/07/2018

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 4 ESP UNIT 2**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10032
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
28/06/2018	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
28/06/2018	ND	ND	0.03

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
28/06/2018	0.000 AC	1.620 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
28/06/2018	0.01 UN

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
28/06/2018	NON-COR	2C	LAVENDER BLUE

®PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
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FIELD SERVICE

DATE	SERVICE
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No.	Unit No.	Serial Number	TC
18.	MAIN TRAFO UNIT 2	1006902	21

A) Analysis

- **Acidity** of 0.008mgKOH/g and **IFT** of 42.1mN/m are **ACCEPTABLE**.
- **Dielectric** of 75kV is **ACCEPTABLE**.
- **Colour, Specific Gravity, and Visualization** for the oil are within the **ACCEPTABLE** range.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 10ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.18% M/dw is graded as **B**. For this transformer, it is advisable to maintain the water content level below 20ppm.
- **Furan Analysis** shows detection of 113ppb of 5M2F. This compound forms as a result of localized and intense overheating of the cellulose involving a high temperature hot spot fault. However, DGA do not show any abnormal profile. Close monitoring is needed to track the speed of degradation proses.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- **Inhibitor content** detected at 0.22% by weight is **ACCEPTABLE**.
- **Corrosive sulfur** test on copper strip resultant in dark orange, thus **Non-corrosive of 1B**.

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B) Recommendation

- Retest **Furan** and **DGA** in 6 months to monitor both the condition and any problems of solid insulation.
- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Other parameters: Do carry-out *Liquid Screen, Liquid Power Factor, Inhibitor, Water Content, Metals-In-Oil, Furan Analysis & DGA periodically for **Condition Monitoring purposes**

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DATE PRINTED 20/07/2018
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - MAIN TRAFU UNIT 2

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
27/11/2017	ONLINE	50	55				
06/12/2017	ONLINE						
22/05/2018	ONLINE			NORMAL		GOOD	NONE
28/06/2018	OFFLINE	50	45				

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 1006902
KVA	- 33000
PRI	- 150000 V
SEC	- 6300 V
KILOGRAMS	- 81000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- OFAF
IMPEDENCE	- 13.03%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 30857

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
27/11/2017	ROUTINE	0.008 AC		70 AC	0.5 AC		CLEAR AC	
06/12/2017	ROUTINE	0.008 AC		67 AC	0.5 AC		CLEAR AC	
22/05/2018	ROUTINE	0.011 AC		46 AC	0.5 AC		CLEAR AC	
28/06/2018	ROUTINE	0.008 AC	42.1 AC	75 AC	0.5 AC	0.8663 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
28/06/2018	19	12733	53386	10	193	2448	ND	2	ND	224	68791

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- YES
H2O COOLED	- NO
OIL PUMPS	- YES
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- YES
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
27/11/2017	37	16	1.04	B
06/12/2017	38	13	0.85	A
22/05/2018	33	14	1.15	B
28/06/2018	28	10	1.18	B

FIELD SERVICE

DATE	SERVICE
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DATE PRINTED 20/07/2018

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - MAIN TRAF0 UNIT 2

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 1006902
KVA	- 33000
PRI	- 150000 V
SEC	- 6300 V
KILOGRAMS	- 81000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- OFAF
IMPEDENCE	- 13.03%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 30857
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- YES
H2O COOLED	- NO
OIL PUMPS	- YES
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- YES
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
06/12/2017	ND	ND	ND	ND	ND	ND
28/06/2018	ND	ND	ND	ND	113	113

RECOMMENDATION:
RETEST 6 MONTHS: THE PRESENCE OF 5M2F INDICATES THE POSSIBILITY OF HOTSPOT
OVERHEATING.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	ALUMINIUM	IRON	COPPER
28/06/2018	ND	ND	0.03

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
28/06/2018	0.000 AC	0.410 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
28/06/2018	0.22 AC

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
28/06/2018	NON-COR	1B	DARK ORANGE

PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
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FIELD SERVICE

DATE	SERVICE
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3.0 Summary of Our Comments & Recommendations:

No.	Sub Name	Unit No.	TC	Recommendation
1.	MAIN TRAFO UNIT 1	100691	4	- Add passivator - Periodic condition monitoring
2.	TRAFO 1 ESP UNIT 1	10937	5	Perform OIL REGENERATION
3.	TRAFO 2 ESP UNIT 1	10035	6	Perform OIL REGENERATION
4.	TRAFO 3 ESP UNIT 1	10031	7	- Add inhibitor 2,6-DBPC and passivator - Periodic condition monitoring
5.	TRAFO 4 ESP UNIT 1	10034	8	Perform OIL REGENERATION
6.	MAIN TRAFO UNIT 3	1122170	9	- Add passivator - Periodic condition monitoring
7.	UAT 3	1035082	10	- Retest 6 months: DGA - Add passivator - Other parameters: Periodic condition monitoring
8.	TRAFO 1 ESP UNIT 3	0073084208	11	- Retest 12 months: Specific Gravity - Add passivator - Other parameters: Periodic condition monitoring
9.	TRAFO 2 ESP UNIT 3	0070084208	12	- Retest 12 months: Specific Gravity - Add passivator - Other parameters: Periodic condition monitoring

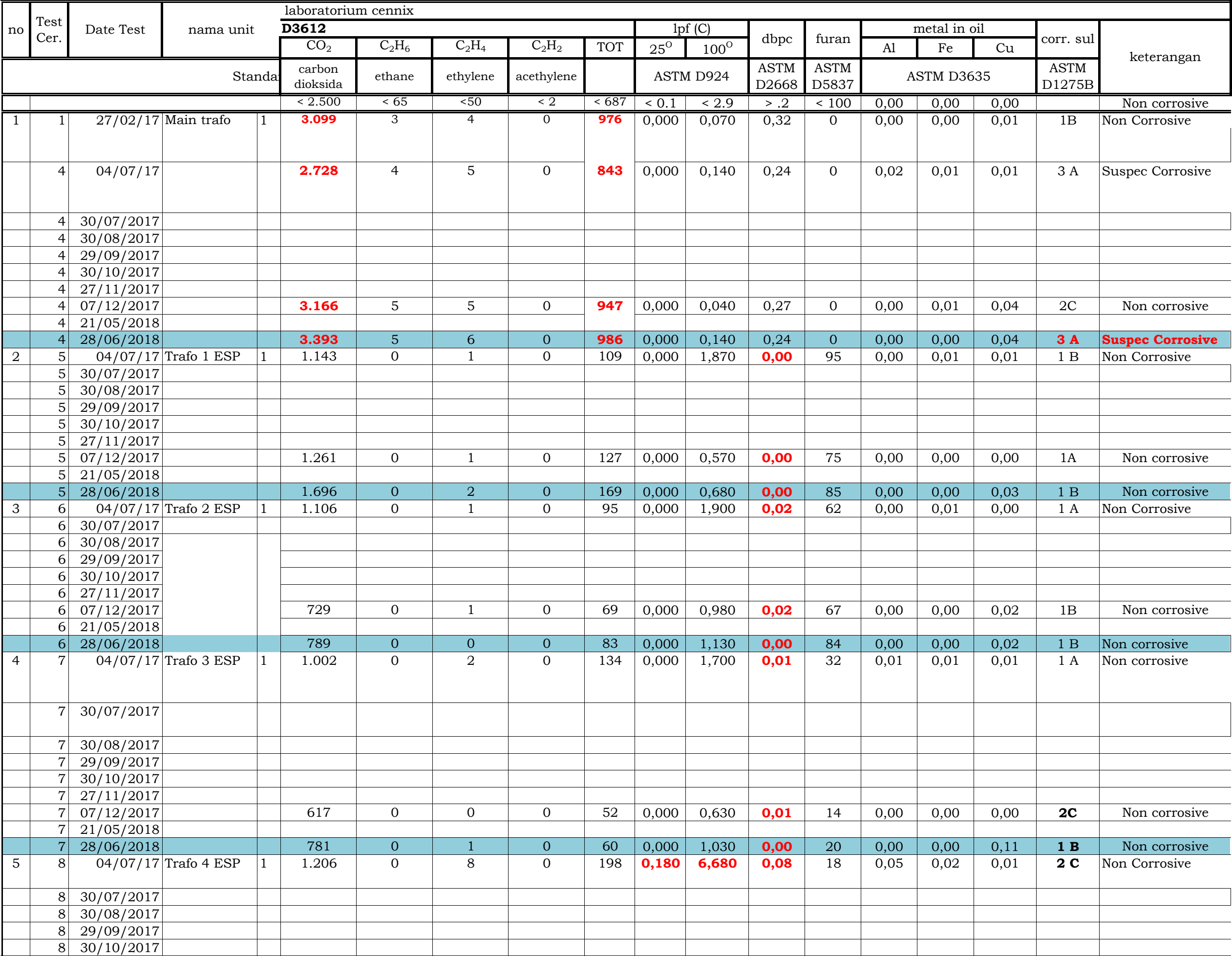
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No.	Sub Name	Unit No.	TC	Recommendation
10.	TRAFO 3 ESP UNIT 3	0072084208	13	- Retest 12 months: Specific Gravity - Retest 6 months: Dielectric and Water Content - Add passivator - Other parameters: Periodic condition monitoring
11.	TRAFO 4 UNIT 3	0071084208	14	- Add inhibitor 2,6-DBPC and passivator - Periodic condition monitoring
12.	ASH HANDLING-1	10820108	15	- Add passivator - Periodic condition monitoring
13.	ASH HANDLING-2	10820107	16	- Retest 6 months: DGA - Add passivator - Other parameters: Periodic condition monitoring
14.	TRAFO 1 ESP UNIT 2	10037	17	- Add inhibitor 2,6-DBPC and passivator - Periodic condition monitoring
15.	TRAFO 2 ESP UNIT 2	10036	18	- Add inhibitor 2,6-DBPC and passivator - Periodic condition monitoring
16.	TRAFO 3 ESP UNIT 2	10033	19	Perform OIL REGENERATION
17.	TRAFO 4 ESP UNIT 2	10032	20	- Add inhibitor 2,6-DBPC and passivator - Periodic condition monitoring
18.	MAIN TRAFO UNIT 2	1006902	21	- Retest 6 months: Furan and DGA - Add passivator - Other parameters: Periodic condition monitoring

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no	Test Cer.	Date Test	nama unit	Serial number	thn opr	Jumlah minyak, liter	Buatan	hasil test														
								acid	ift	diel.	color	spec. Graf.	visual	sedi.	moisture		dga ASTM					
															ppm	grade	H ₂	O ₂	N ₂	CH ₄	CO	
Standard yang dipakai								ASTM D974	ASTM D971	IEC 156	ASTM D1500	ASTM D4052	ASTM D1524	ASTM D1698	ASTM D1533	hydrogen	oksigen	nitrogen	methane	carbon monoksida		
	BATASAN OPERASI							< 0,06	> 32	> 30	< 3,5	> 0,84	C		< 20		< 100			< 120	< 350	
1	1	27/02/17	Main trafo	1	1006901	2011	30.857	Wolong	0,014	39,9	99	0,5	0,8693	C		9	A	135	15.834	57.625	27	807
	4	04/07/17							0,017	41,5	58	0,5	0,8691	C		10	A	147	9.471	4.328	32	655
	4	30/07/2017							0,017		70	0,5		C		8	B					
	4	30/08/2017							0,017		92	0,5		C		6	A					
	4	29/09/2017							0,014		82	0,5		C		8	A					
	4	30/10/2017							0,014		82	0,5		C		6	A					
	4	27/11/2017							0,014		53	0,5		C		8	A					
	4	07/12/2017							0,011	40,1	87	0,5	0,8745	C		7	A	183	12.895	43.794	39	715
	4	21/05/2018							0,011		77	0,5		C		10	B					
	4	28/06/2018							0,014	41,9	69	0,5	0,8716	C		7	A	176	11.044	46.776	41	758
2	5	04/07/17	Trafo 1 ESP	1	10937	2010	1.143	Fujia	0,056	25,3	50	1,5	0,8529	C		18	D	35	20.494	67.487	4	69
	5	30/07/2017							0,059		57	1,5		C		20	D					
	5	30/08/2017							0,056		65	1,5		C		18	D					
	5	29/09/2017							0,059		58	1,5		C		19	D					
	5	30/10/2017							0,059		23	1,5		C		32	C					
	5	27/11/2017							0,062		25	1,5		C		33	C					
	5	07/12/2017							0,062	25,0	49	1,5	0,8593	C		18	D	39	19.933	67.198	4	83
	5	21/05/2018							0,065		38	1,5		C		33	E					
	5	28/06/2018							0,067	25,9	32	1,5	0,8560	C		30	E	44	20.039	64.529	4	119
3	6	04/07/17	Trafo 2 ESP	1	10035	2010	1.143	Fujia	0,042	26,6	41	1,5	0,8540	C		21	D	27	18.639	66.781	3	64
	6	30/07/2017							0,045		25	1,5		C		27	E					
	6	30/08/2017							0,042		42	1,5		C		14	C					
	6	29/09/2017							0,048		59	1,5		C		19	D					
	6	30/10/2017							0,051		62	1,5		C		17	B					
	6	27/11/2017							0,051		50	1,5		C		18	B					
	6	07/12/2017							0,063	27,3	51	1,5	0,8589	C		17	D	20	19.360	61.590	3	45
	6	21/05/2018							0,056		45	1,5		C		20	D					
	6	28/06/2018							0,059	27,6	55	1,5	0,8570	C		17	C	22	22.494	64.443	4	57
4	7	04/07/17	Trafo 3 ESP	1	10031	2010	1.143	Fujia	0,048	28,4	44	1,0	0,8537	C		32	E	32	15.273	65.111	5	95
	7	30/07/2017							0,042		77	1,5				10	D					
	7	30/08/2017							0,042		61	1,5		C		13	C					
	7	29/09/2017							0,045		72	1,5		C		14	B					
	7	30/10/2017							0,048		72	1,5		C		16	B					
	7	27/11/2017							0,051		62	1,5		C		17	C					
	7	07/12/2017							0,063	29,3	57	1,5	0,8624	C		16	C	7	14.730	65.738	3	42
	7	21/05/2018							0,056		48	1,5		C		21	D					
	7	28/06/2018							0,053	29,7	62	1,5	0,8605	C		17	D	9	17.958	66.007	4	46
5	8	04/07/17	Trafo 4 ESP	1	10034	2010	1.143	Fujia	0,028	26,3	24	1,5	0,8596	C		29	E	13	12.265	66.770	5	172
	8	30/07/2017							0,022		88	1,5		C		15	D					
	8	30/08/2017							0,028		44	1,5		C		13	C					
	8	29/09/2017							0,031		53	1,0		C		15	B					
	8	30/10/2017							0,028		48	1,0		C		18	B					



no	Test Cer.	Date Test	nama unit	MALAYSIAN LAB RECOMMENDATION		PAK SOETJAHJONO (TXM ADVISOR) RECOMMENDATION
Standar						
1	1	27/02/17	Main trafo	1	Retest 6 months: DGA Check leakages Other parameters: Periodic Condition Monitoring	FOR TEST 28 Juni 2018 Masih ada beberapa trafo ESP (no 1 Unit 1 dan no 2 Unit 2) yang Acid dan IFT di bawah standard. Kami sarankan untuk ganti minyak . Trafo-trafo ESP (no 2,3,4 unit 1 , no 2,3,4 Unit 2 , n 2,3 Unit 3) dan Ash Handling 1 yang Grade Misture di bawah B, kami sarankan untuk dilakukan Purifikasi Pada ESP 2,3,4 Unit 3 ditemukan Spesific Grafity di bawah stadanrd. Kemungkinan penggantian oli menggunakan bukan Mineral Oli . Mohon dicek kembali type oli pengganti atau dilakukan trending dalam 12 bulan mendatang. FURAN Semua Trafo ESP Unit 1, ESP 2 Unit 2 dan ESP 3 Unit 3 telah ditemukan jejak Furan akibat moisture tinggi yang cukup lama. Kami sarankan untuk menjaga kondisi moisture pada waktu selanjutnya agar perkembangan Furan tidak berlanjut Main Trafo Unit 2 ditemukan jejak Furan 5M2F (ada hot spot) tetapi tidak ditemukan Combustible Gas berlebih pada test DGA. Hal ini mengakibatkan pengamanan terhadapat gangguan petir kurang memadai. Kami sarankan untuk diperbaiki baiksystem groundingnya maupun arresternya CORROSIVE SULPHUR Main Trafo Unit 1 dan 3 serta trafo UAT dideteksi mulai ada corrosive sulphur. Untuk mencegah percepatan pertumbuhannya dan memperpanjang umur trafo, kami sarankan menambah Passivator
	4	04/07/17			Retest 6 months: DGA Add inhibitor 2,6-DBPC and Passivator Other parameters: Periodic Condition Monitoring	
	4	30/07/2017			Periodic condition monitoring	
	4	30/08/2017			Periodic condition monitoring	
	4	29/09/2017			Periodic condition monitoring	
	4	30/10/2017			Periodic condition monitoring	
	4	27/11/2017			Periodic condition monitoring	
	4	07/12/2017			Retest 6 mnth : DGA	
	4	21/05/2018			Periodic condition monitoring	
	4	28/06/2018			Add passivator	
2	5	04/07/17	Trafo 1 ESP	1	Perform OIL REGENERATION	FOR TEST 21 Mei 2018 Meningat banyak trafo ESP (Tc 5,6,7,8,11,12,13) yang kurang memenuhi syarat. Kami sarankan untuk diganti minyak saja. Selain jumlah minyaknya sedikit biaya juga akan jauh lebih murah, Saran kami saat penggantian minyak melalui purifikasi dengan temperatur lebih dari 650C dan vacuum diatas 74 CMHg diatas 7 cycle Trafo ESP (Tc 6,7,8) Furan (FOL) banyak berkurang karena moisture cukup banyak dalam kertas dan kemungkinan juga kualitas kertas kurang bagus Untuk trafo Ash Handling sebaiknya segera dipurifikasi juga dengan temperatur diats 650c dan vacuum diatas 74 CMHg sekitar >7 cycle Trafo ESP 1 unit 2 , nilai Dielectric nya cukup rendah ditambah dengan kondisi Moisture yang buruk. sebaiknya segera dilakukan
	5	30/07/2017			Perform OIL REGENERATION	
	5	30/08/2017			Perform OIL REGENERATION	
	5	29/09/2017			Perform OIL REGENERATION	
	5	30/10/2017			Perform OIL REGENERATION	
	5	27/11/2017			Perform OIL REGENERATION	
	5	07/12/2017			Perform OIL REGENERATION	
	5	21/05/2018			Perform OIL REGENERATION	
	5	28/06/2018			Perform OIL REGENERATION	
3	6	04/07/17	Trafo 2 ESP	1	Perform OIL REGENERATION	
	6	30/07/2017			Perform OIL REGENERATION	
	6	30/08/2017			Perform OIL REGENERATION	Retest 6 months: Acidity, IFT and Liquid Power Factor Other parameters: Periodic Condition Monitoring
	6	29/09/2017			Perform OIL REGENERATION	
	6	30/10/2017			Perform OIL REGENERATION	
	6	27/11/2017			Perform OIL REGENERATION	
	6	07/12/2017			Perform OIL REGENERATION	
	6	21/05/2018			Perform OIL REGENERATION	
	6	28/06/2018			Perform OIL REGENERATION	
4	7	04/07/17	Trafo 3 ESP	1	Retest 6 months: Acidity, IFT, LPF, Diel and Water Content Add inhibitor 2,6-DBPC and Passivator Other parameters: Periodic Condition Monitoring	
	7	30/07/2017			Retest 6 months: Acidity, IFT and Liquid Power Factor Other parameters: Periodic Condition Monitoring	
	7	30/08/2017			Periodic condition monitoring	
	7	29/09/2017			Periodic condition monitoring	Retest 6 months: Acidity, IFT, Dielectric and Water Content Add inhibitor 2,6-DBPC and PASSIVATOR
	7	30/10/2017			Periodic condition monitoring	
	7	27/11/2017			Periodic condition monitoring	
	7	07/12/2017			Retest 6 months: Acidity, IFT	
	7	21/05/2018			Retest 6 months: Acidity, IFT, Dielectric and Water Content	
	7	28/06/2018			Add inhibitor 2,6-DBPC and PASSIVATOR	
5	8	04/07/17	Trafo 4 ESP	1	Perform OIL REGENERATION Inspect silica gel	
	8	30/07/2017			Perform OIL REGENERATION	
	8	30/08/2017			Perform OIL REGENERATION	
	8	29/09/2017			Perform OIL REGENERATION	
	8	30/10/2017			Perform OIL REGENERATION	



no	Test Cer.	Date Test	nama unit	Serial number	thn opr	Jumlah minyak, liter	Buatan	hasil test														
								acid	ift	diel.	color	spec. Graf.	visual	sedi.	moisture		dga ASTM					
															ppm	grade	H ₂	O ₂	N ₂	CH ₄	CO	
Standard yang dipakai								ASTM D974	ASTM D971	IEC 156	ASTM D1500	ASTM D4052	ASTM D1524	ASTM D1698	ASTM D1533	hydrogen	oksigen	nitrogen	methane	carbon monoksida		
			BATASAN OPERASI					< 0,06	> 32	> 30	< 3,5	> 0,84	C		< 20		< 100			< 120	< 350	
	8	27/11/2017						0,034		41	1,5		C		30	D						
	8	07/12/2017						0,031	27,2	41	1,5	0,8645	C		18	D	9	13.331	63.527	3	70	
	8	21/05/2018						0,034		37	1,5		C		23	D						
	8	28/06/2018						0,037	27,5	55	1,5	0,8653	C		20	D	9	17.962	64.248	4	75	
6	2	27/02/17	Main trafo	3	1122170	2010	26.706	Shandong	0,017	39,7	87	0,5	0,8781	C		9	A	4	13.330	43.140	9	808
	9	04/07/17						0,017	42,3	78	0,5	0,8780	C		15	B	24	10.131	44.084	11	641	
	9	30/07/2017						0,014		77	0,5		C		17	B						
	9	30/08/2017						0,017		86	0,5		C		8	A						
	9	29/09/2017						0,014		86	0,5		C		10	A						
	9	30/10/2017						0,014		88	0,5		C		9	A						
	9	27/11/2017						0,014		87	0,5		C		8	A						
	9	07/12/2017						0,014	41,5	84	0,5	0,8932	C			A	39	14.023	43.226	11	668	
	9	21/05/2018						0,014		58	0,5		C		11	B						
		28/06/2018						0,014	41,0	85	0,5	0,8805	C		8	A	18	11.431	48.283	13	700	
7	3	27/02/17	Trafo UAT	3	1035082	2010	4.749	Shandong	0,017	37,6	94	0,5	0,8768	C		9	A	20	12.535	56.879	20	985
	10	04/07/17						0,022	38,1	60	0,5	0,8767	C		11	A	20	16.028	68.308	20	971	
	10	30/07/2017						0,025		88	0,5		C		14	B						
	10	30/08/2017						0,022		89	0,5		C		11	A						
	10	29/09/2017						0,020		80	0,5		C		12	A						
	10	30/10/2017						0,022		86	0,5		C		11	A						
	10	27/11/2017						0,022		86	0,5		C		12	A						
	10	07/12/2017						0,020	37,6	86	0,5	0,8200	C		12	A	36	18.752	66.758	1	1.116	
	10	21/05/2018						0,020		81	0,5		C		14	B						
	10	28/06/2018						0,022	37,5	79	0,5	0,8793	C		10	A	35	13.798	68.180	22	1.144	
8	11	04/07/17	Trafo 1 ESP	3	007308428	2010	731	Zhejian	0,022	23,2	32	1,5	0,8432	C		26	D	14	12.943	68.807	3	62
	11	30/07/2017						0,028		47	1,5		C		21	D						
	11	30/08/2017						0,028		36	1,5		C		22	D						
	11	29/09/2017						0,025		45	1,5		C		20	C						
	11	30/10/2017						0,025		40	1,5		C		19	D						
	11	27/11/2017						0,025		31	1,5		C		34	C						
	11	07/12/2017						0,022	34,6	31	1,5	0,8190	C		31	C	14	12.739	65.097	3	66	
	11	21/05/2018						0,022		31	0,5		C		27	D						
	11	28/06/2018						0,022	35,0	60	0,5	0,8154	C		10	B	19	15.552	64.448	3	57	
9	12	04/07/17	Trafo 2 ESP	3	007008428	2010	731	Zhejian	0,118	16,2	24	3,0	0,8435	C		39	D	38	16.781	70.523	5	129
	12	30/07/2017						0,112		28	3,0		C		38	E						
	12	30/08/2017						0,109		34	2,5		C		17	D						
	12	29/09/2017						0,115		21	2,5		C		37	D						
	12	30/10/2017						0,118		33	2,5		C		20	D						
	12	27/11/2017						0,017		25	0,5		C		28	B						
	12	07/12/2017						0,022	29,7	29	0,1	0,8106	C		44	D	42	19.523	66.059	3	102	
	12	21/05/2018						0,022		28	0,5		C		30	E						
	12	28/06/2018						0,025	30,2	34	0,5	0,8080	C		16	C	43	22.272	68.719	4	38	



no	Test Cer.	Date Test	nama unit	laboratorium cennix														keterangan
				D3612					lpf (C)		dbpc	furan	metal in oil			corr. sul		
				CO ₂	C ₂ H ₆	C ₂ H ₄	C ₂ H ₂	TOT	25 ^o	100 ^o			Al	Fe	Cu			
Standar				carbon dioksida	ethane	ethylene	acethylene		ASTM D924		ASTM D2668	ASTM D5837	ASTM D3635			ASTM D1275B		
					< 2.500	< 65	<50	< 2	< 687	< 0.1	< 2.9	> .2	< 100	0,00	0,00	0,00		Non corrosive
	8	27/11/2017																
	8	07/12/2017			720	0	1	0	83	0,010	2,180	0,06	0	0,00	0,00	0,02	2C	Non corrosive
	8	21/05/2018																
	8	28/06/2018			1.054	0	2	0	90	0,060	2,850	0,02	21	0,00	0,00	0,05	3 A	Suspec Corrosive
6	2	27/02/17	Main trafo	3	2.878	0	1	0	822	0,000	0,090	0,30	0	0,00	0,03	0,02	1A	Non corrosive
	9	04/07/17			2.250	1	1	0	678	0,000	0,130	0,22	0	0,01	0,01	0,02	3 A	Suspec Corrosive
	9	30/07/2017																
	9	30/08/2017																
	9	29/09/2017																
	9	30/10/2017																
	9	27/11/2017																
	9	07/12/2017			2.456	0	1	0	719	0,000	0,070	0,22	0	0,01	0,00	0,03	2C	Non corrosive
	9	21/05/2018																
		28/06/2018			1.512	1	1	0	733	0,000	0,110	0,19	0	0,00	0,01	0,06	3 A	Suspec Corrosive
7	3	27/02/17	Trafo UAT	3	3.877	6	7	0	1.038	0,000	0,100	0,30	0	0,00	0,01	0,04	1A	Non Corrosive
	10	04/07/17			3.536	5	7	0	1.023	0,000	0,390	0,23	0	0,02	0,01	0,04	3 A	Suspec Corrosive
	10	30/07/2017																
	10	30/08/2017																
	10	29/09/2017																
	10	30/10/2017																
	10	27/11/2017																
	10	07/12/2017			3.622	5	7	0	1.165	0,000	0,170	0,23	0	0,00	0,01	0,08	2C	Non corrosive
	10	21/05/2018																
	10	28/06/2018			3.644	5	7	1	1.214	0,000	0,140	0,21	0	0,00	0,01	0,11	3 A	Suspec Corrosive
8	11	04/07/17	Trafo 1 ESP	3	1.243	0	0	0	79	0,000	0,350	0,00	0	0,01	0,01	0,00	1 B	Non corrosive
	11	30/07/2017																
	11	30/08/2017																
	11	29/09/2017																
	11	30/10/2017																
	11	27/11/2017																
	11	07/12/2017			612	0	0	0	83	0,000	0,040	0,18	0	.01	0,00	0,00	2E	Non corrosive
	11	21/05/2018																
	11	28/06/2018			1.391	0	0	0	79	0,000	0,050	0,16	0	0,00	0,00	0,02	1 B	Non corrosive
9	12	04/07/17	Trafo 2 ESP	3	2.431	3	5	0	180	0,000	1,330	0,00	169	0,00	0,01	0,01	2 C	Non corrosive
	12	30/07/2017																
	12	30/08/2017																
	12	29/09/2017																
	12	30/10/2017																
	12	27/11/2017																
	12	07/12/2017			912	0	0	0	147	0,000	0,030	0,20	0	0,00	0,00	0,00	3A	Non corrosive
	12	21/05/2018																
	12	28/06/2018			1.226	0	0	0	85	0,000	0,040	0,19	0	0,00	0,00	0,02	3 A	Suspec Corrosive



no	Test Cer.	Date Test	nama unit	MALAYSIAN LAB RECOMMENDATION		PAK SOETJAHJONO (TXM ADVISOR) RECOMMENDATION
Standar						
	8	27/11/2017			Perform OIL REGENERATION	Kondisi semua trafo ESP Unit 2 (Tc 17, 18, 19, 20) Grade Moisture rendah, yang artinya kandungan air dalam isolasi kertas cukup tinggi. Kami sarankan untuk dimonitor dengan test DGA agar bisa diantisipasi jika ada kondisi gangguan thermal yang terjadi pada trafo trafo tersebut.
	8	07/12/2017			Perform OIL REGENERATION	
	8	21/05/2018			Perform OIL REGENERATION	
	8	28/06/2018			Perform OIL REGENERATION	
6	2	27/02/17	Main trafo	3	Retest 6 months: DGA	
					Other parameters: Periodic Condition Monitoring	
	9	04/07/17			Add Passivator	
					Periodic Condition Monitoring	
	9	30/07/2017			Periodic condition monitoring	
	9	30/08/2017			Periodic condition monitoring	
	9	29/09/2017			Periodic condition monitoring	
	9	30/10/2017			Periodic condition monitoring	
	9	27/11/2017			Periodic condition monitoring	
	9	07/12/2017			Periodic condition monitoring	
	9	21/05/2018			Periodic condition monitoring	
		28/06/2018			Add passivator	
7	3	27/02/17	Trafo UAT	3	Retest 6 months: DGA	
					Other parameters: Periodic Condition Monitoring	
	10	04/07/17			Add Passivator	
					Periodic Condition Monitoring	
					Inspect silica gel	
	10	30/07/2017			Periodic condition monitoring	
	10	30/08/2017			Periodic condition monitoring	
	10	29/09/2017			Periodic condition monitoring	
	10	30/10/2017			Periodic condition monitoring	
	10	27/11/2017			Periodic condition monitoring	
	10	07/12/2017			Periodic condition monitoring	
	10	21/05/2018			Periodic condition monitoring	
	10	28/06/2018			Retest 6 months: DGA	
					Add Passivator	
8	11	04/07/17	Trafo 1 ESP	3	Perform OIL REGENERATION	
	11	30/07/2017			Perform OIL REGENERATION	
	11	30/08/2017			Perform OIL REGENERATION	
	11	29/09/2017			Perform OIL REGENERATION	
	11	30/10/2017			Perform OIL REGENERATION	
	11	27/11/2017			Perform OIL REGENERATION	
	11	07/12/2017			Perform OIL PURIFICATION	
	11	21/05/2018			Perform OIL PURIFICATION	
	11	28/06/2018			Retest 12 months: Specific Gravity	
					Add Passivator	
9	12	04/07/17	Trafo 2 ESP	3	Perform OIL REGENERATION	
	12	30/07/2017			Perform OIL REGENERATION	
	12	30/08/2017			Perform OIL REGENERATION	
	12	29/09/2017			Perform OIL REGENERATION	
	12	30/10/2017			Perform OIL REGENERATION	
	12	27/11/2017			Perform OIL REGENERATION	
	12	07/12/2017			Perform OIL REGENERATION	
	12	21/05/2018			Perform OIL PURIFICATION	
	12	28/06/2018			Retest 12 months: Specific Gravity	
					Add Passivator	



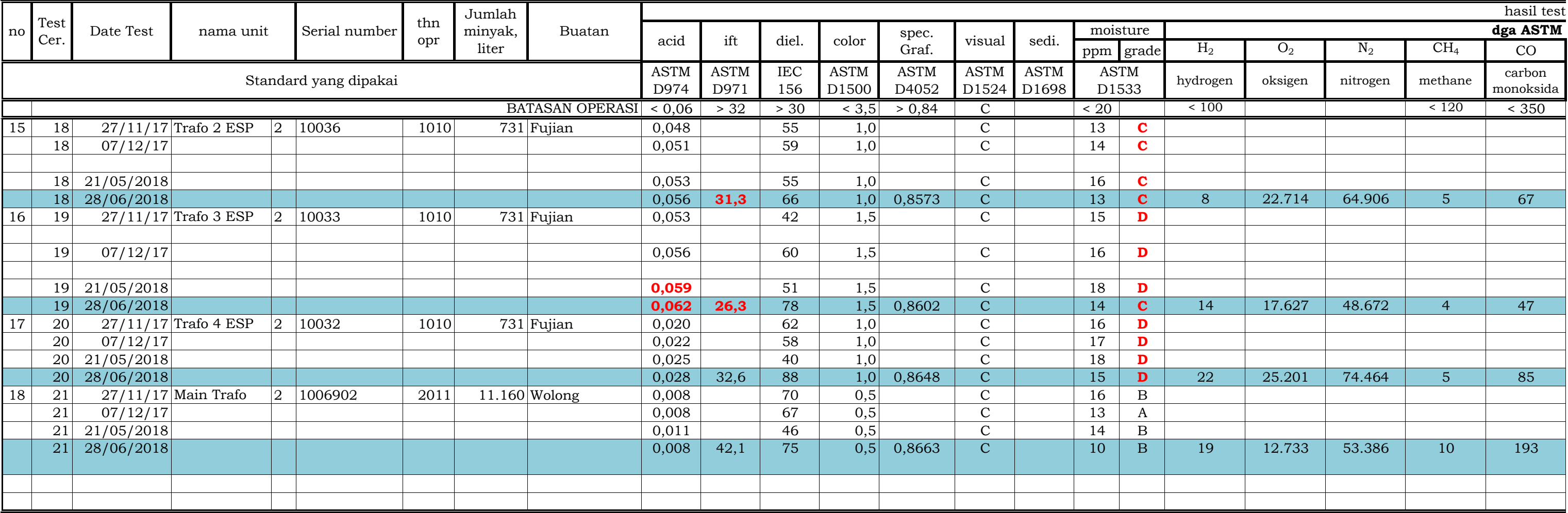
no	Test Cer.	Date Test	nama unit	Serial number	thn opr	Jumlah minyak, liter	Buatan	hasil test													
								acid	ift	diel.	color	spec. Graf.	visual	sedi.	moisture		dga ASTM				
															ppm	grade	H ₂	O ₂	N ₂	CH ₄	CO
Standard yang dipakai								ASTM D974	ASTM D971	IEC 156	ASTM D1500	ASTM D4052	ASTM D1524	ASTM D1698	ASTM D1533	hydrogen	oksigen	nitrogen	methane	carbon monoksida	
			BATASAN OPERASI					< 0,06	> 32	> 30	< 3,5	> 0,84	C		< 20		< 100			< 120	< 350
10	13	04/07/17	Trafo 3 ESP	3	007208428	2010	731 Zhejian	0,025	27,8	40	1,5	0,8415	C		27	D	21	16.364	70.694	3	38
	13	30/07/2017						0,025		66	1,5		C		18	D					
	13	30/08/2017						0,025		50	1,5		C		21	C					
	13	29/09/2017						0,028		59	1,5		C		22	C					
	13	30/10/2017						0,028		55	1,5		C		28	D					
	13	27/11/2017						0,014		39	0,5		C		24	B					
	13	07/12/2017						0,014	39,8	45	0,5	0,8099	C		29	B	15	14.515	65.277	2	40
	13	21/05/2018						0,014		54	0,5		C		23	D					
	13	28/06/2018						0,017	38,4	43	0,5	0,8075	C		22	C	30	15.202	64.188	4	65
11	14	04/07/17	Trafo 4 ESP	3	007108428	2010	731 Zhejian	0,020	31,2	62	1,0	0,8420	C		19	D	10	18.669	66.036	3	24
	14	30/07/2017						0,022		60	1,0		C		15	C					
	14	30/08/2017						0,020		55	1,0		C		17	D					
	14	29/09/2017						0,022		59	1,0		C		15	C					
	14	30/10/2017						0,025		55	1,0		C		15	D					
	14	27/11/2017						0,022		69	1,0		C		16	C					
	14	07/12/2017						0,025	29,8	43	1,0	0,8471	C		19	D	14	15.414	63.712	2	22
	14	21/05/2018						0,028		37	1,0		C		21	D					
	14	28/06/2018						0,025	30,5	71	1,0	0,8442	C		11	A	20	14.690	67.822	2	22
12	15	04/07/17	Ash Handling		10820108	2011	1.023 Wolong	0,022	38,1	38	1,0	0,8779	C		28	C	35	10.984	71.058	20	441
	15	30/07/2017						0,025		56	1,0		C		21	D					
	15	30/08/2017						0,022		53	1,0		C		16	B					
	15	29/09/2017						0,022		57	1,0		C		16	B					
	15	30/10/2017						0,022		45	1,0		C		20	B					
	15	27/11/2017						0,020		36	1,0		C		29	C					
	15	07/12/2017						0,017	37,5	34	1,0	0,8932	C		26	B	49	12.034	67.124	20	558
	15	21/05/2018						0,022		33	1,0		C		26	C					
	15	28/06/2018						0,025	37,5	42	1,0	0,8804	C		24	C	53	10.939	69.394	20	557
13	16	04/07/17	Ash Handling		10820107	2011	1.023 Wolong	0,022	37,3	36	1,0	0,8788	C		21	C	18	21.157	64.224	4	129
	16	30/07/2017						0,022		40	1,0		C		32	D					
	16	30/08/2017						0,022		31	1,0		C		29	C					
	16	29/09/2017						0,022		28	1,5		C		31	B					
	16	30/10/2017						0,020		46	1,5		C		21	C					
	16	27/11/2017						0,022		47	1,5		C		20	C					
	16	07/12/2017						0,020	36,6	24	1,5	0,8940	C		32	D	20	15.791	63.530	5	275
	16	21/05/2018						0,020		24	1,5		C		32	D					
	16	28/06/2018						0,022	35,0	50	1,5	0,8811	C		16	B	32	22.904	67.139	4	249
14	17	27/11/17	Trafo 1 ESP	2	10037	1010	731 Fujian	0,020		48	1,0		C		21	D					
	17	07/12/17						0,022		22	1,0		C		35	E					
	17	21/05/2018						0,025		24	1,0		C		32	E					
	17	28/06/2018						0,025	32,9	77	1,0	0,8693	C		11	B	93	17.225	43.231	5	57

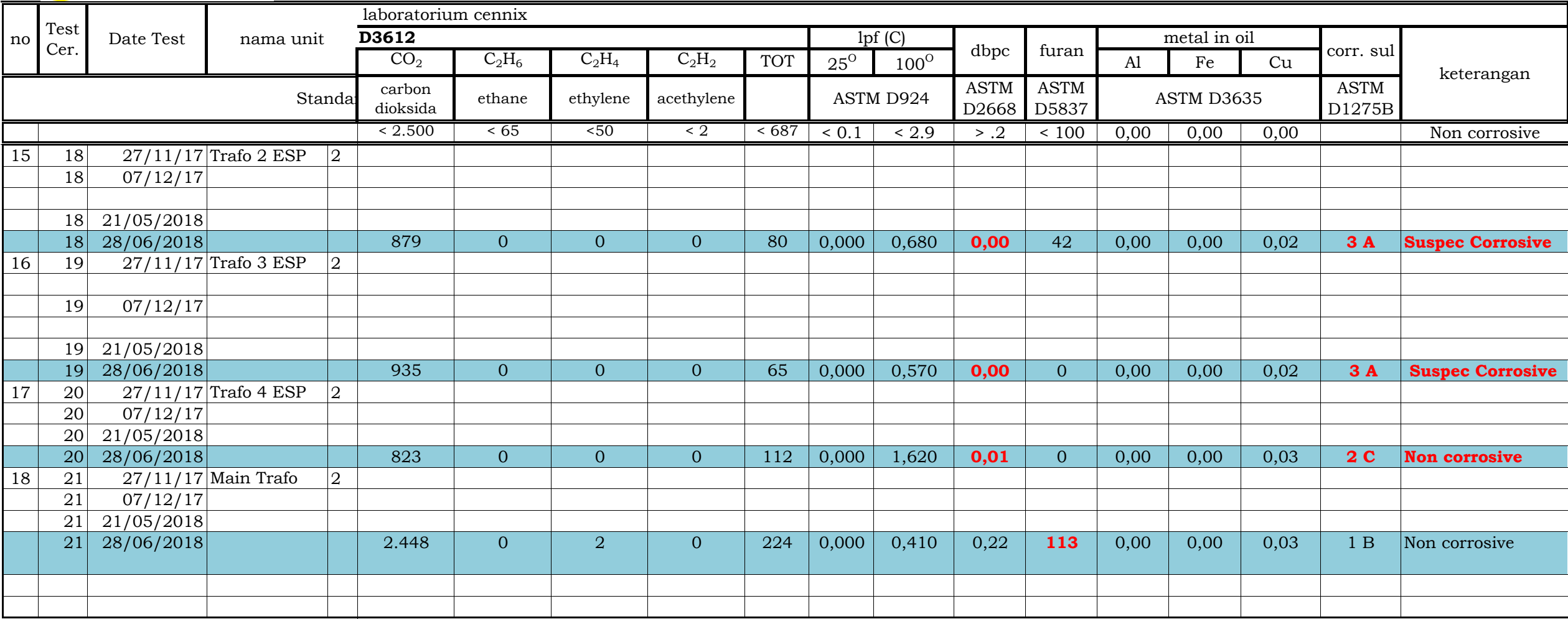


no	Test Cer.	Date Test	nama unit	laboratorium cennix														
				D3612					lpf (C)		dbpc	furan	metal in oil			corr. sul	keterangan	
				CO ₂	C ₂ H ₆	C ₂ H ₄	C ₂ H ₂	TOT	25 ^o	100 ^o			Al	Fe	Cu			
Standar				carbon dioksida	ethane	ethylene	acethylene		ASTM D924		ASTM D2668	ASTM D5837	ASTM D3635			ASTM D1275B		
					< 2.500	< 65	<50	< 2	< 687	< 0.1	< 2.9	> .2	< 100	0,00	0,00	0,00		Non corrosive
10	13	04/07/17	Trafo 3 ESP	3	1.068	0	0	0	62	0,000	0,620	0,01	0	0,00	0,01	0,00	2 C	Non corrosive
	13	30/07/2017																
	13	30/08/2017																
	13	29/09/2017																
	13	30/10/2017																
	13	27/11/2017																
	13	07/12/2017			559	0	0	0	57	0,000	.08	.21	0	0,00	0,00	0,00	2E	Non corrosive
	13	21/05/2018																
	13	28/06/2018			1.833	0	0	0	99	0,000	0,010	0,20	4	0,00	0,00	0,02	2 E	Non corrosive
11	14	04/07/17	Trafo 4 ESP	3	665	0	0	0	37	0,000	1,360	0,01	0	0,00	0,00	0,00	3 A	Suspec Corrosive
	14	30/07/2017																
	14	30/08/2017																
	14	29/09/2017																
	14	30/10/2017																
	14	27/11/2017																
	14	07/12/2017			675	0	0	0	38	0,000	0,370	0,03	0	0,00	0,00	0,00	1B	Non corrosive
	14	21/05/2018																
	14	28/06/2018			683	0	0	0	44	0,000	0,550	0,00	0	0,00	0,03	0,02	1 B	Non corrosive
12	15	04/07/17	Ash Handling		5.037	7	8	0	511	0,000	0,730	0,23	0	0,00	0,01	0,01	1 B	Non corrosive
	15	30/07/2017																
	15	30/08/2017																
	15	29/09/2017																
	15	30/10/2017																
	15	27/11/2017																
	15	07/12/2017			5.440	8	9	0	644	0,000	0,240	0,24	0	0,00	0,00	0,01	2C	Non corrosive
	15	21/05/2018																
	15	28/06/2018			5.425	9	8	0	647	0,000	0,460	0,22	0	0,00	0,01	0,04	1 B	Non corrosive
13	16	04/07/17	Ash Handling		1.676	0	1	0	152	0,000	1,350	0,22	0	0,00	0,01	0,00	3 A	Suspec Corrosive
	16	30/07/2017																
	16	30/08/2017																
	16	29/09/2017																
	16	30/10/2017																
	16	27/11/2017																
	16	07/12/2017			2.783	0	4	0	304	0,000	0,270	0,24	3	0,00	0,00	0,01	1A	Non corrosive
	16	21/05/2018																
	16	28/06/2018			3.112	0	5	0	290	0,000	0,490	0,20	0	0,00	0,00	0,02	3 A	Suspec Corrosive
14	17	27/11/17	Trafo 1 ESP	2														
	17	07/12/17																
	17	21/05/2018																
	17	28/06/2018			775	0	2	0	157	0,030	3,420	0,00	0	0,00	0,06	0,03	2 C	Non corrosive



no	Test Cer.	Date Test	nama unit	MALAYSIAN LAB RECOMMENDATION		PAK SOETJAHJONO (TXM ADVISOR) RECOMMENDATION
Standards						
10	13	04/07/17	Trafo 3 ESP	3	Perform OIL REGENERATION	
	13	30/07/2017			Perform OIL REGENERATION	
	13	30/08/2017			Perform OIL REGENERATION	
	13	29/09/2017			Perform OIL REGENERATION	
	13	30/10/2017			Perform OIL REGENERATION	
	13	27/11/2017			Perform OIL REGENERATION	
	13	07/12/2017			Perform OIL PURIFICATION	
	13	21/05/2018			Perform OIL PURIFICATION	
	13	28/06/2018			Retest 12 months: Specific Gravity Retest 6 months: Dielectric and Water Content Add Passivator	
11	14	04/07/17	Trafo 4 ESP	3	Retest 6 months: Acidity, IFT and LPF Add inhibitor 2,6-DBPC and Passivator Other parameters: Periodic Condition Monitoring	
	14	30/07/2017			Retest 6 months: Acidity, IFT and Liquid Power Factor Other parameters: Periodic Condition Monitoring	
	14	30/08/2017			Periodic condition monitoring	
	14	29/09/2017			Periodic condition monitoring	
	14	30/10/2017			Periodic condition monitoring	
	14	27/11/2017			Periodic condition monitoring	
	14	07/12/2017			Retest 6 months: Acidity, IFT	
	14	21/05/2018			Retest 6 months: Dielectric and Water Content	
	14	28/06/2018			Add inhibitor 2,6-DBPC and PASSIVATOR	
12	15	04/07/17	Ash Handling		Add Passivator Periodic Condition Monitoring	
	15	30/07/2017			Periodic condition monitoring	
	15	30/08/2017			Periodic condition monitoring	
	15	29/09/2017			Periodic condition monitoring	
	15	30/10/2017			Periodic condition monitoring	
	15	27/11/2017			Periodic condition monitoring	
	15	07/12/2017			Periodic condition monitoring	
	15	21/05/2018			Periodic condition monitoring	
	15	28/06/2018			Add passivator	
13	16	04/07/17	Ash Handling		Add Passivator Periodic Condition Monitoring	
	16	30/07/2017			Retest 6 months: Dielectric and Water Content Other parameters: Periodic Condition Monitoring	
	16	30/08/2017			Periodic Condition Monitoring	
	16	29/09/2017			Retest 6 months: Dielectric and Water Content	
	16	30/10/2017			Retest 6 months: Dielectric and Water Content	
	16	27/11/2017			Periodic Condition Monitoring	
	16	07/12/2017			Perform OIL PURIFICATION	
	16	21/05/2018			Perform OIL PURIFICATION	
	16	28/06/2018			Retest 6 months: DGA Add Passivator	
14	17	27/11/17	Trafo 1 ESP	2	Retest 6 months: Dielectric and Water Content	
					Other parameters: Periodic Condition Monitoring	
	17	07/12/17			OIL PURIFICATION	
	17	21/05/2018			OIL PURIFICATION	
	17	28/06/2018			Add inhibitor 2,6-DBPC and PASSIVATOR	







no	Test Cer.	Date Test	nama unit	MALAYSIAN LAB RECOMMENDATION		PAK SOETJAHJONO (TXM ADVISOR) RECOMMENDATION
Standar						
15	18	27/11/17	Trafo 2 ESP	2	Periodic condition monitoring	
	18	07/12/17			Retest 6 months: Acidity and IFT	
					Periodic condition monitoring	
	18	21/05/2018			Retest 6 months: Acidity and IFT	
	18	28/06/2018			Add inhibitor 2,6-DBPC and PASSIVATOR	
16	19	27/11/17	Trafo 3 ESP	2	Retest 6 months: Acidity and IFT	
					Other parameters: Periodic Condition Monitoring	
	19	07/12/17			Retest 6 months: Acidity and IFT	
					Other parameters: Periodic Condition Monitoring	
	19	21/05/2018			Retest 6 months: Acidity and IFT	
	19	28/06/2018			Perform OIL REGENERATION	
17	20	27/11/17	Trafo 4 ESP	2	Periodic condition monitoring	
	20	07/12/17			Periodic condition monitoring	
	20	21/05/2018			Periodic condition monitoring	
	20	28/06/2018			Add inhibitor 2,6-DBPC and PASSIVATOR	
18	21	27/11/17	Main Trafo	2	Periodic condition monitoring	
	21	07/12/17			Periodic condition monitoring	
	21	21/05/2018			Periodic condition monitoring	
	21	28/06/2018			Retest 6 months: Furan and DGA Add Passivator	

Summary by Soetjahjono
TxM Indonesia